

ON3VZ

Measuring as a radio amateur with the NanoVNA-H4 and the tinySA

Step by step, for anyone who has never used a VNA or a spectrum
analyser

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on3vz.github.io

How to use this manual. Every figure shows a reconstructed screen with red numbered markers; the table below it explains each number. For every action you will find not only what to do, but exactly how to do it. Menu names differ slightly between firmware versions: see appendix F.

The companion document *SWR sweep per band* turns chapter 8 into a fixed routine for every HF band. Both are available in English and Dutch at on3vz.github.io.

Contents

0. Before you start

- 0.1 How to read a menu path
- 0.2 The four safety rules
- 0.3 Discharging the coax: this is how
- 0.4 Disconnecting the transceiver
- 0.5 The three phases

1. What these two instruments do

- 1.1 The NanoVNA-H4 in plain language
- 1.2 S11 and S21: the two kinds of measurement
- 1.3 The tinySA in plain language
- 1.4 Which instrument for which question
- 1.5 Frequency coverage and your bands

2. Kit list and connecting up

- 2.1 Phase 1: measuring antennas
- 2.2 Phase 2: measuring your transmitter with the tinySA
- 2.3 Phase 3: chokes and ferrites
- 2.4 Tightening connectors correctly

3. Operating the NanoVNA and calibrating it

- 3.1 First power-up
- 3.2 Reading the screen
- 3.3 The main menu
- 3.4 Setting the frequency range
- 3.5 Choosing traces, format and channel
- 3.6 Calibration: the single most important step
- 3.7 Loading a saved calibration
- 3.8 Memory plan (a suggestion)
- 3.9 Using the marker

4. Reading the graphs

- 4.1 Which format for which question
- 4.2 SWR
- 4.3 LOGMAG: return loss
- 4.4 R and X: resistance and reactance
- 4.5 Phase
- 4.6 Several traces at once
- 4.7 LOGMAG on CH1: transmission

5. The Smith chart, demystified

- 5.1 Why a Smith chart?

- 5.2 The chart, built up in four steps
- 5.3 Reading the marker values at the top
- 5.4 Your own antenna on the Smith chart
- 5.5 Exercise: tuning an antenna tuner with the Smith chart

6. Phase 1: measuring antennas

- 6.1 SWR and resonance of an HF antenna
- 6.2 A multiband antenna in one view
- 6.3 Too long or too short: Smith and R/X together
- 6.4 Measuring in the shack or at the antenna
- 6.5 VHF and UHF antennas via N

7. Phase 1: coax and filters

- 7.1 Coax length and fault finding (TDR)
- 7.2 Measuring a filter (S21)

8. The NanoVNA on your PC

- 8.1 Connecting and recognising the NanoVNA
- 8.2 Downloading NanoVNA-Saver and connecting
- 8.3 The window, panel by panel
- 8.4 Calibrating in the software
- 8.5 Comparing before and after
- 8.6 Cable length in the software (TDR)
- 8.7 Saving for your log or blog
- 8.8 Updating the firmware (only if needed)

9. Phase 2: the tinySA and your transmitter

- 9.1 First power-up
- 9.2 Why you always use external attenuation
- 9.3 Connecting
- 9.4 Measuring harmonics
- 9.5 Checking frequency and power
- 9.6 Hunting interference sources

10. Phase 3: chokes and ferrites

- 10.1 What are you measuring?
- 10.2 Connecting
- 10.3 Calibrating and measuring
- 10.4 Converting to ohms
- 10.5 Measuring clip-on ferrites on a cable

Appendices

- A. Quick card: measuring an antenna
- B. Start and stop per band
- C. Conversion tables

- D. Glossary
- E. Links and sources
- F. Menu names per firmware

0. Before you start

This guide teaches you to measure with two small instruments in three phases. You do not need any prior knowledge. For every action you will find not only what to do, but also exactly how to do it.

0.1 How to read a menu path

A menu path such as `CALIBRATE > RESET` means:

- Step 1 Open the main menu.** Tap anywhere on the screen with a stylus or the supplied plectrum, or press the jog switch on top of the unit straight down.
- Step 2 A column of buttons appears on the right.** Tap `CALIBRATE`, or push the jog switch sideways until `CALIBRATE` lights up green, then press it in.
- Step 3 A new column appears.** Tap `RESET`.
- Step 4 To go back one level,** tap `← BACK`. To close the menu completely, tap the graph next to the menu, or push the jog switch to the left a few times.

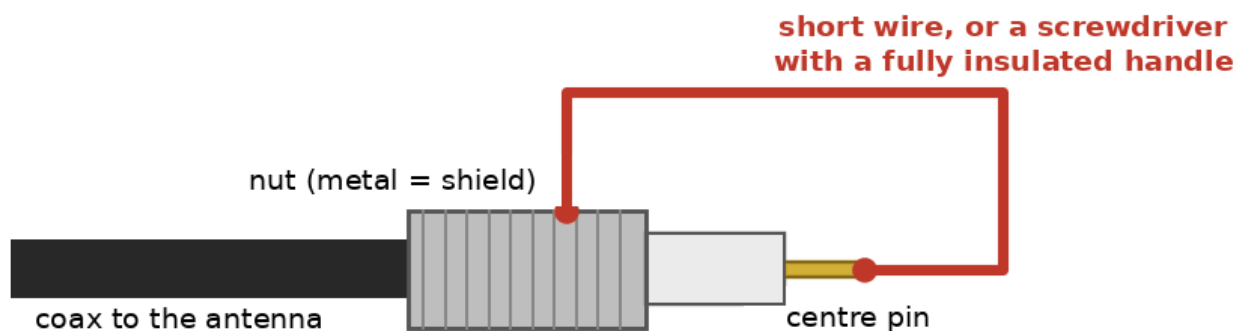
0.2 The four safety rules

SAFETY

1. **Never transmit into the NanoVNA or the tinySA.** Never connect a transceiver, amplifier or handheld directly. Even 1 watt permanently destroys the input.
2. **With the tinySA, always attenuate a transmitter signal completely with external attenuators** (chapter 9). Never count the tinySA's internal attenuator as protection.
3. **Discharge every outdoor antenna before connecting it**, exactly as described in 0.3.
4. **Disconnect the transceiver from the antenna line before connecting the VNA**, so that nobody can transmit by accident. Never measure when there is a thunderstorm anywhere nearby.

0.3 Discharging the coax: this is how

An outdoor wire antenna (EFHW, dipole, longwire) builds up static charge from wind, dry air or a distant thunderstorm. That charge sits on the centre pin of your coax. Connect the NanoVNA and it discharges straight into the sensitive input. So you bleed that charge off to the shield yourself first.



Hold the centre pin against the metal nut for 2 seconds

Figure 1 · Bridge the centre pin and the metal nut of the PL-259 for two seconds.

WHAT YOU NEED

- A piece of copper wire about 10 cm long with 1 cm stripped at both ends, **or** a screwdriver with a fully insulated handle.
- Better and permanent: a **PL-259 shorting plug** (a PL-259 in which the centre pin and the body are joined with a short wire), or a **50 ohm dummy load** with a PL connector.

STEPS

Step 1 Make sure the transceiver is no longer connected to the coax (see 0.4).

Step 2 Hold the antenna coax's PL-259 by the cable or the plastic, **not** by the pin in the middle.

Step 3 Place one end of the wire (or the metal shaft of the screwdriver) against the metal nut of the PL-259.

Step 4 Touch the centre pin with the other end and hold it there for 2 seconds.

Step 5 If you hear or feel a small tick, or see a tiny spark, there was charge on the line. That is normal, and exactly why you do this.

Step 6 Connect the NanoVNA within a minute (see 2.4). If you wait longer, or it is windy, discharge again.

TIP

A shorting plug makes this even easier: screw it onto the PL-259 for 2 seconds and take it off again. Keep it in the bag with your NanoVNA.

0.4 Disconnecting the transceiver

Step 1 Switch the transceiver off.

Step 2 Find the coax that runs from your antenna switch to the transceiver.

- Step 3** Turn the nut of the PL-259 on the antenna switch side anticlockwise until it spins freely, and pull the plug straight out.
- Step 4** Set the antenna switch to the antenna you want to measure. The NanoVNA will go on the free socket where the transceiver was.

0.5 The three phases

You do not need to buy or learn everything at once. The guide is built in three phases; phase 1 alone already answers most antenna questions.

Phase	What you can do	Chapters
Phase 1 (start here)	Measure antennas: SWR, resonance, impedance, coax, filters	1 to 8
Phase 2	Measure your own transmitter with the tinySA: harmonics, frequency, power, and hunting interference	9
Phase 3	Test chokes and ferrites (common-mode current)	10

1. What these two instruments do

1.1 The NanoVNA-H4 in plain language

A VNA (vector network analyser) sends out a very weak test signal of its own and measures what happens to it: does it come back (**reflection**), or does it arrive at the other side (**transmission**)? From that it calculates SWR, impedance, loss and even the length of a cable. The H4 covers 50 kHz to 1.5 GHz and has a 4 inch screen.

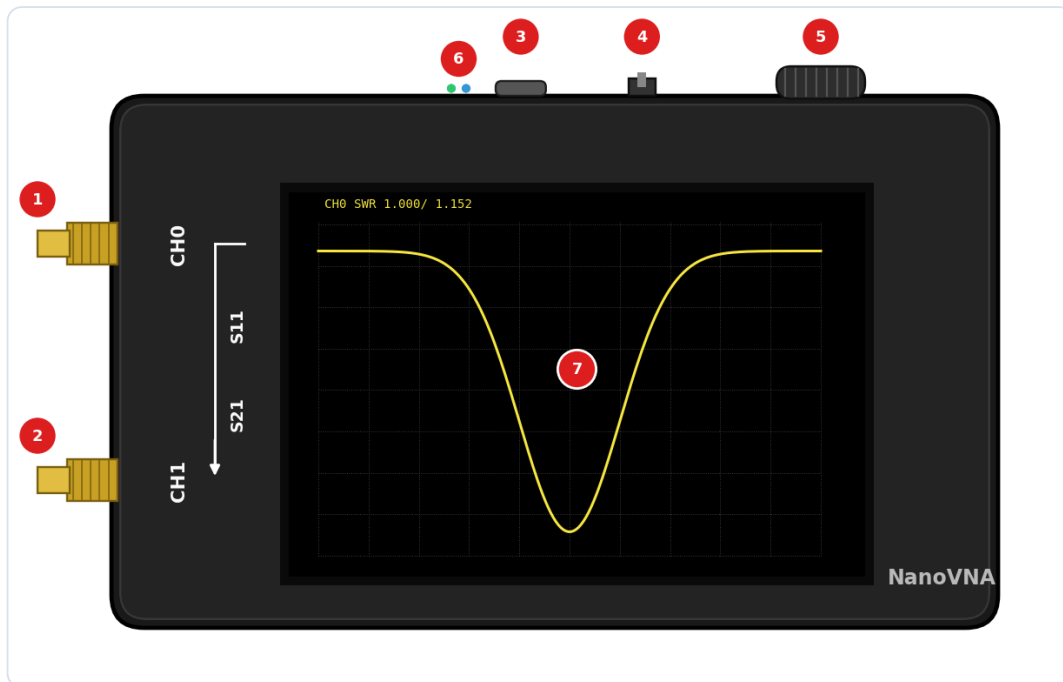


Figure 2 · The NanoVNA-H4 and its connections.

No.	What you see here
1	CH0 or Port 1: this is where the antenna goes. This port sends the test signal and measures what comes back (S11).
2	CH1 or Port 2: only for transmission measurements (filters, chokes). It receives whatever passes through the device under test (S21).
3	USB-C: charging and connection to the PC.
4	On/off slide switch.
5	Jog switch: push left or right to select or move the marker, press straight down to confirm or open the menu.
6	LEDs: the battery LED flashes while charging and stays on when the battery is full.
7	Touchscreen. It is pressure-sensitive (resistive): tap with a fingernail, a stylus or the supplied plectrum, not with a light fingertip.

1.2 S11 and S21: the two kinds of measurement

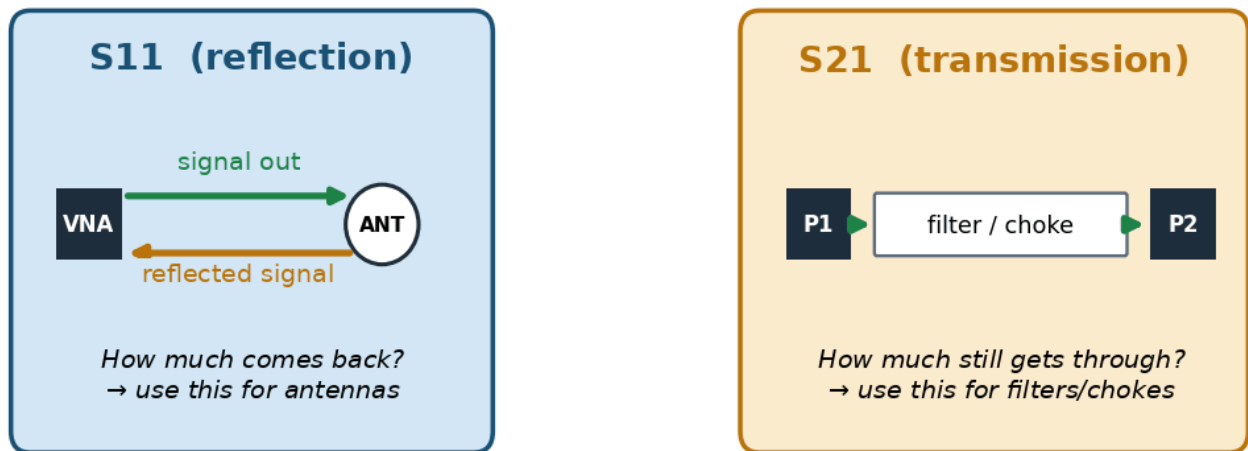


Figure 3 · S11 measures what bounces back, S21 measures what gets through.

Think of shouting at a wall. **S11** is the echo that comes back to you. **S21** is what someone on the other side of the wall can still hear.

- **S11** (reflection, always on CHO) is what you use for antennas. Little echo means the antenna accepts the signal well: low SWR.
- **S21** (transmission, from CHO to CH1) is what you use for filters and chokes. You see how many dB are lost on the way.

1.3 The tinySA in plain language

The tinySA does the opposite: it transmits nothing, it listens. It is a **spectrum analyser**. It shows which frequencies are present in a signal and how strong each one is. That is how you see whether your transmitter is also radiating unwanted signals at twice or three times your frequency (harmonics), or where an interference source is.



Figure 4 · The tinySA (basic model).

No.	What you see here
1	LOW input (0.1 to 350 MHz): for HF, 6 m and 2 m. Maximum +10 dBm.
2	HIGH input (240 to 960 MHz): for 70 cm. Less accurate. Maximum +10 dBm.
3	USB-C.
4	On/off.
5	Jog switch , works the same as on the NanoVNA.
6	Screen: frequency horizontally, signal strength in dBm vertically.

1.4 Which instrument for which question

Your question	Instrument	Section
Is my antenna tuned correctly?	NanoVNA	6.1
Is my antenna too long or too short?	NanoVNA	5, 6.3
Do all bands of my multiband antenna work?	NanoVNA	6.2
How long is my coax, and is there a fault in it?	NanoVNA	7.1
How much does my low-pass filter attenuate?	NanoVNA	7.2
Is my QMX radiating unwanted harmonics?	tinySA	9.4
Where is that interference on 40 m coming from?	tinySA	9.6
Does my choke or ferrite work on 40 m?	NanoVNA	10

1.5 Frequency coverage and your bands

Band	NanoVNA-H4	tinySA	Connector
160 m to 10 m	excellent	LOW input, excellent	PL (SO-239 adapter)
6 m	excellent	LOW input, excellent	PL or N
2 m	very good	LOW input, good	N adapter
70 cm	good	HIGH input, limited	N adapter
23 cm	usable	not possible	N adapter

2. Kit list and connecting up

The links go straight to the product page. Prices are rough orders of magnitude, including VAT, as seen at the time of writing: shops change prices and stock all the time, so always check before ordering.

2.1 Phase 1: measuring antennas

Item	What it is for	Approx. price	Shop
NanoVNA-H4	The instrument itself. In the box: 2 SMA cables, open/short/load calibration caps, an SMA female-female coupler and a USB-C cable	about €57	Eleshop
SMA female to SO-239 female adapter	Goes on the end of the SMA cable; the PL-259 of your HF coax screws onto it	about €5	Eleshop
N female to SMA male adapter	For VHF/UHF coax that ends in an N plug	about €3	Eleshop

Phase 1 total: roughly €65, plus shipping. Useful to have (you probably do already): a PL-259 shorting plug or a piece of wire for discharging (0.3).

2.2 Phase 2: measuring your transmitter with the tinySA

Item	What it is for	Approx. price	Shop
40 dB attenuator, 10 W, with heat sink	First stage, directly after the transmitter. It absorbs the power	about €49	Eleshop
10 dB attenuator, 2 W	Second stage. Brings the signal far below the tinySA's limit	about €10	Eleshop
BNC male to SMA female adapter	From the QMX's BNC output to the attenuator	about €5	Eleshop
PL-PL patch cable, 90 cm (optional)	Only for measuring the IC-7300 at 5 W, together with the SO-239 adapter	about €12	Deltron

2.3 Phase 3: chokes and ferrites

Item	What it is for	Approx. price	Shop
BNC male cable with crocodile clips, 2 pieces	Red clip on the choke, black clips joined together	about €20	Eleshop
BNC female to SMA male adapter, 2 pieces	Connects the clip leads to CHO and CH1	about €6	Eleshop

2.4 Tightening connectors correctly



Figure 5 · SMA cable on CHO, SO-239 adapter on its end, the coax's PL-259 on the adapter.

- Step 1 SMA cable on CHO:** hold the cable straight in front of the port and turn only the small hexagonal nut clockwise, with your fingers. Do not twist the body of the cable along with it. Stop as soon as the nut will not go any further.
- Step 2 SO-239 adapter on the other end of the SMA cable:** same method. Hold the adapter still and tighten the cable's nut.
- Step 3 The coax's PL-259 on the adapter:** push the plug straight onto the adapter until the centre pin is fully in, then turn the knurled nut clockwise until it is hand-tight.
- Step 4 Disconnecting** is the same in reverse order, always anticlockwise.

CAUTION

Never use pliers on SMA connectors, and never screw a heavy coax or adapter directly onto the instrument's SMA port. The supplied SMA cable in between takes up the strain.

3. Operating the NanoVNA and calibrating it

3.1 First power-up

- Step 1

Charge the unit. Plug the USB-C cable into the NanoVNA and into a phone charger (5 V) or your PC. The battery LED flashes while charging and stays on once it is full (allow 1 to 2 hours).
- Step 2

Switch it on with the slide switch on top.
- Step 3

Check the firmware version: `CONFIG > VERSION`. Write the version number down. Press the jog switch to go back.
- Step 4

Calibrate the touchscreen: `CONFIG > TOUCH CAL`. The screen asks you to tap the top left corner, then the bottom right. Tap precisely in the corner with the tip of the plectrum.
- Step 5

Test it with `CONFIG > TOUCH TEST`: draw a line with the plectrum. The line must appear exactly under the tip. If it does not, repeat TOUCH CAL. Press the jog switch to stop.
- Step 6

Save with `CONFIG > SAVE`. Without this step the touch calibration is lost when you switch off.

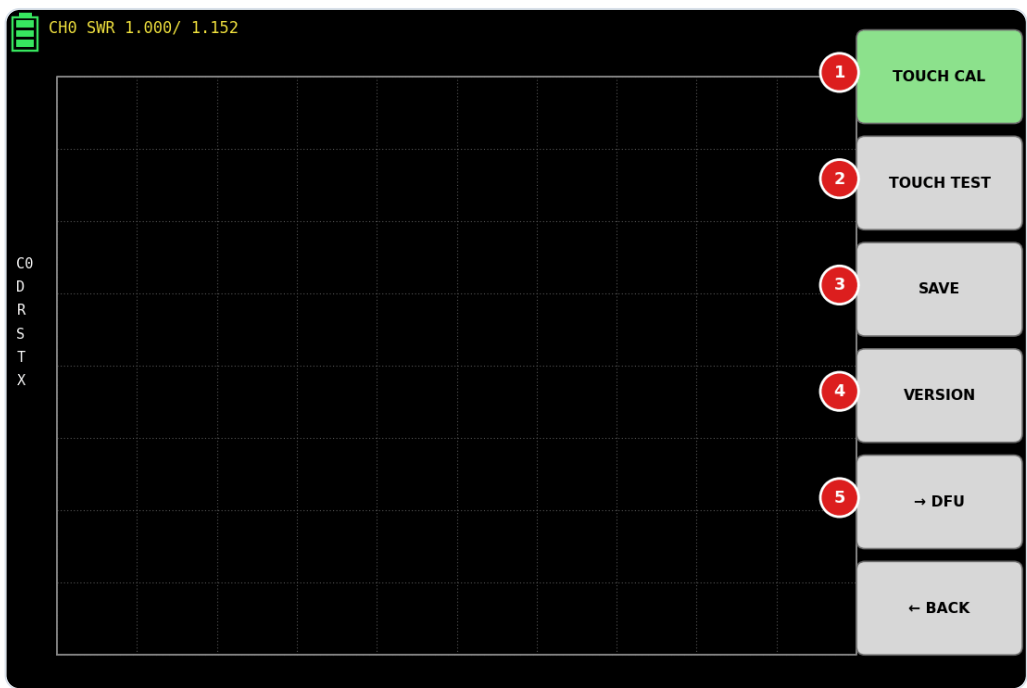


Figure 6 · The CONFIG menu.

No.	What you see here
1	<code>TOUCH CAL</code> : calibrate the touchscreen.
2	<code>TOUCH TEST</code> : check that taps land where they should.
3	<code>SAVE</code> : store the instrument settings.
4	<code>VERSION</code> : show the firmware version.
5	<code>DFU</code> : firmware update mode (only for updates, see 8.8).

3.2 Reading the screen

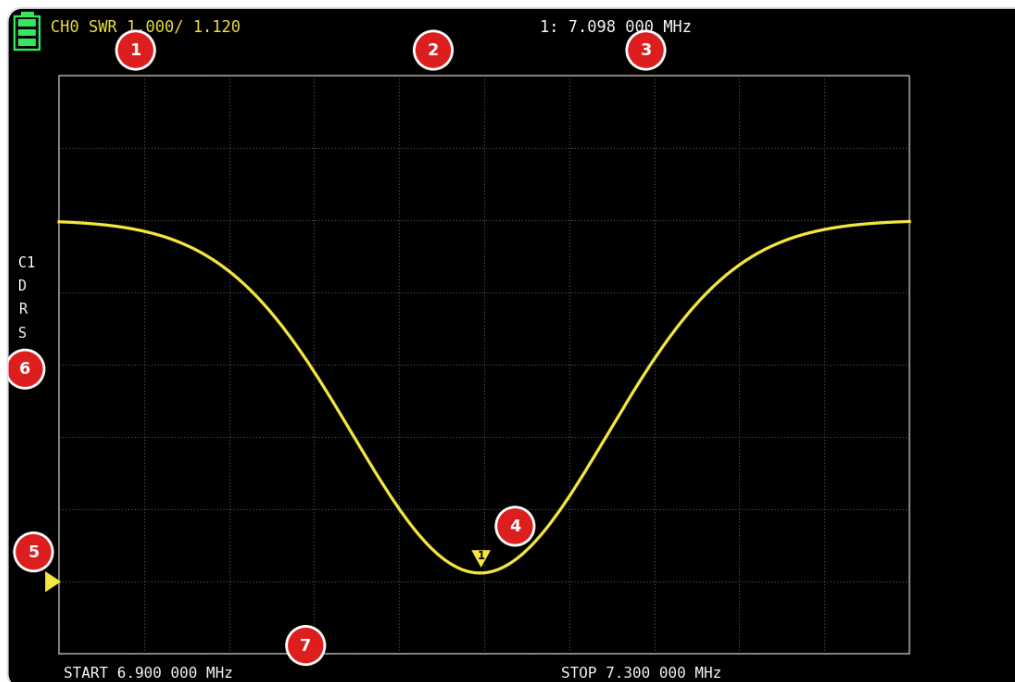


Figure 7 · An SWR measurement of a 40 m antenna, every element numbered.

No.	What you see here
1	Trace header: channel (CHO), format (SWR), scale (1.000 per division) and the value at the marker. Its colour matches the colour of the line.
2	The value at the active marker: SWR 1.12.
3	Frequency of marker 1: 7.098 MHz.
4	The marker: the small numbered triangle on the line. Move it by pushing the jog switch left or right, or by dragging the triangle with your stylus.
5	Reference line (the triangle on the left): this is where SWR 1.0 sits. The closer the curve comes to it, the better.
6	Calibration status. C1 means the calibration from memory slot 1 is active. If nothing is shown here, the unit is not calibrated and the measurement cannot be trusted.
7	Start and stop frequency: the screen shows everything between these two frequencies.

3.3 The main menu

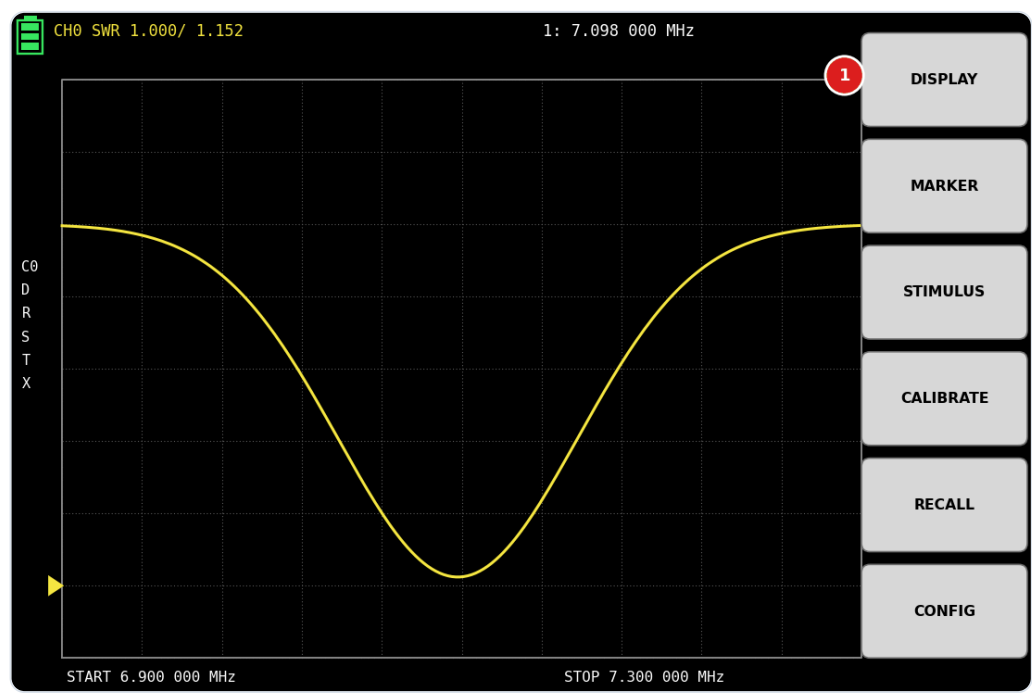


Figure 8 · The main menu on the right of the screen.

Menu	What it is for	Most important submenus
DISPLAY	What you see	TRACE, FORMAT, SCALE, CHANNEL, TRANSFORM
MARKER	Reading off points	SELECT MARKER, SEARCH
STIMULUS	Frequency range	START, STOP, CENTER, SPAN, SWEEP POINTS
CALIBRATE (or CAL)	Calibrating	CALIBRATE, SAVE, RESET
RECALL	Load a saved setup	RECALL 0 to 4
CONFIG	Instrument settings	TOUCH CAL, TOUCH TEST, SAVE, VERSION, DFU

3.4 Setting the frequency range

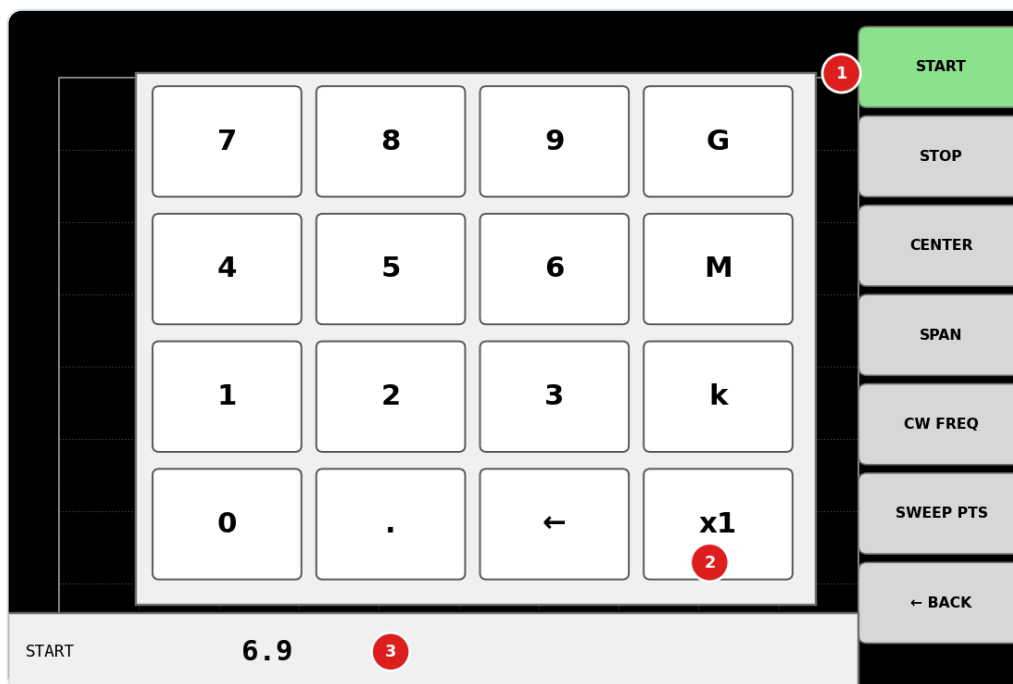


Figure 9 · Entering a start frequency of 6.9 MHz.

No.	What you see here
1	The STIMULUS menu, with START selected.
2	Unit key: G = GHz, M = MHz, k = kHz, x1 = Hz. Tapping a unit ends the entry immediately.
3	The value you are typing. Made a typo? The ← key deletes the last digit.

Step 1	Tap STIMULUS > START . A keypad appears.
Step 2	Tap 6 , then the decimal point, then 9 , and finally M . Bottom left now reads START 6.900 000 MHz.
Step 3	Tap STIMULUS > STOP , then 7 . 3 M .
Step 4	For more measuring points: STIMULUS > SWEEP POINTS > 201 . More points give a finer curve but a slower sweep.

TIP

Do not make the range too wide. With 101 points across 1 to 30 MHz the unit only measures every 290 kHz, and a sharp dip can fall right between two points. Per band, about 400 kHz is enough (appendix B).

3.5 Choosing traces, format and channel

A **trace** is one line on the screen. The NanoVNA can show four at once, each in its own colour, and each trace has its own format (how it is drawn) and channel (which port it reads).

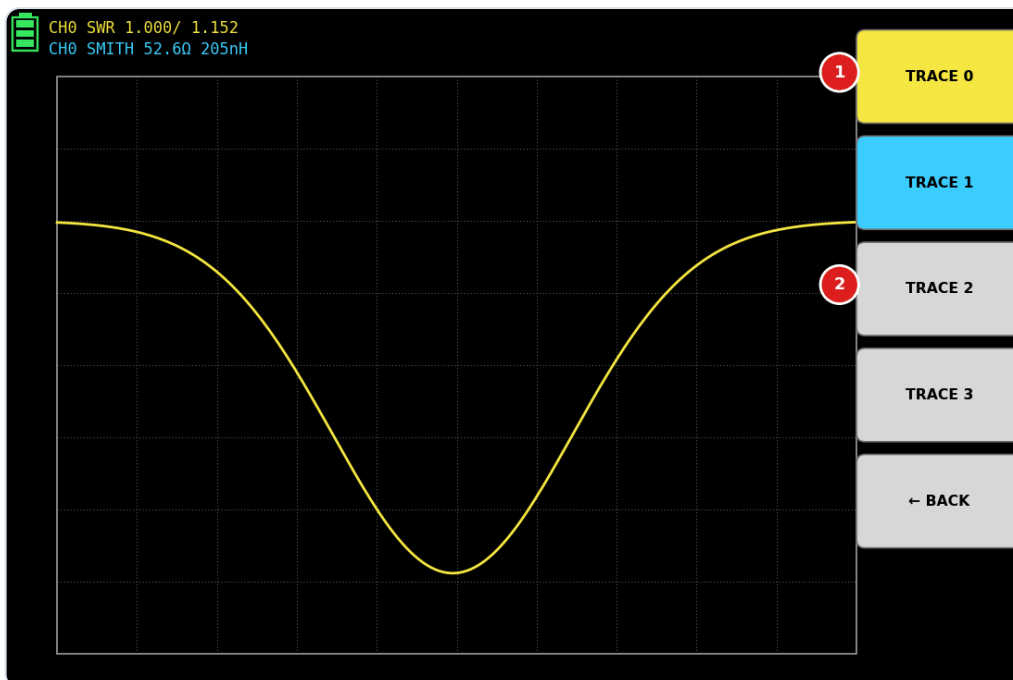


Figure 10 · The TRACE menu. Coloured buttons are switched on.

No.	What you see here
1	Coloured button = the trace is on. One tap makes it the active trace (the one you adjust next).
2	Grey button = the trace is off. Tap to switch it on. To switch an active trace off, tap it once more.

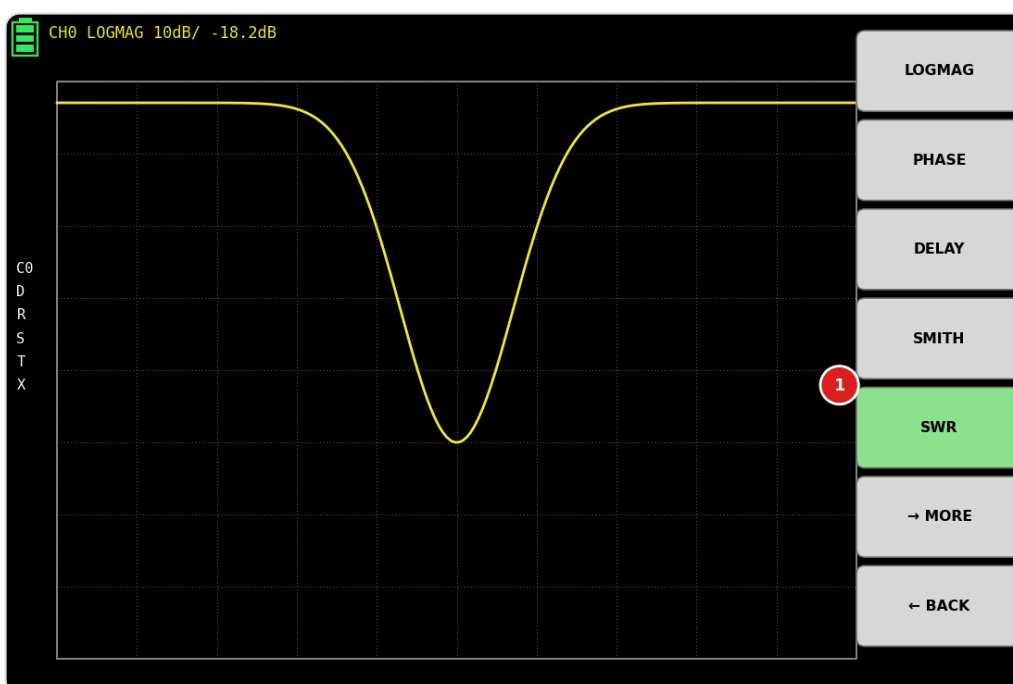


Figure 11 · The FORMAT menu, with SWR selected.

Step 1 Choose the trace you want to change: `DISPLAY > TRACE > TRACE 0`. At the top of the screen, a small triangle now appears in front of trace 0's line: that is the active trace.

- Step 2** Tap **← BACK** and choose the format: **FORMAT > SWR** for antennas, **SMITH** for impedance, **LOGMAG** for filters and chokes.
- Step 3** Choose the channel: **DISPLAY > CHANNEL > CH0 REFLECT** for antennas, **CH1 THROUGH** for filters and chokes.
- Step 4** Choose the scale: **DISPLAY > SCALE > SCALE/DIV**, type **1** and **x1** for SWR, or **10** and **x1** for LOGMAG.
- Step 5** Choose the reference line: **DISPLAY > SCALE > REFERENCE POSITION**, type **1** and **x1** for SWR (second grid line from the bottom), or **7** and **x1** for LOGMAG (near the top).
- Step 6** Repeat for trace 1 if you want a second line, for example the Smith chart next to the SWR.

3.6 Calibration: the single most important step

Calibration teaches the NanoVNA what a "perfect open", a "perfect short" and a "perfect 50 ohm load" look like **at the end of your cable**. From then on it automatically subtracts the influence of that cable from every measurement. Without calibration, you are mostly measuring your own test cable.

RECOGNISING THE THREE CAPS

Cap	How to recognise it	What it simulates
OPEN	Only white insulating material inside, no metal pin in the middle. Sometimes marked O	A cable connected to nothing
SHORT	All metal inside, pin and rim joined. Sometimes marked S	A short circuit
LOAD	A small pin inside, often silver rather than gold. Sometimes marked L or 50 Ω	A perfect 50 ohm load

GOLDEN RULE

Calibrate at the point where the device under test will be connected. For an antenna that is the end of the SMA cable. You can add the small SO-239 adapter afterwards: on HF its effect is negligible.

CAUTION

A calibration is only valid for the frequency range in which you made it. If you change START or STOP, calibrate again or load a saved calibration.

STEP A: CLEAR THE OLD CALIBRATION

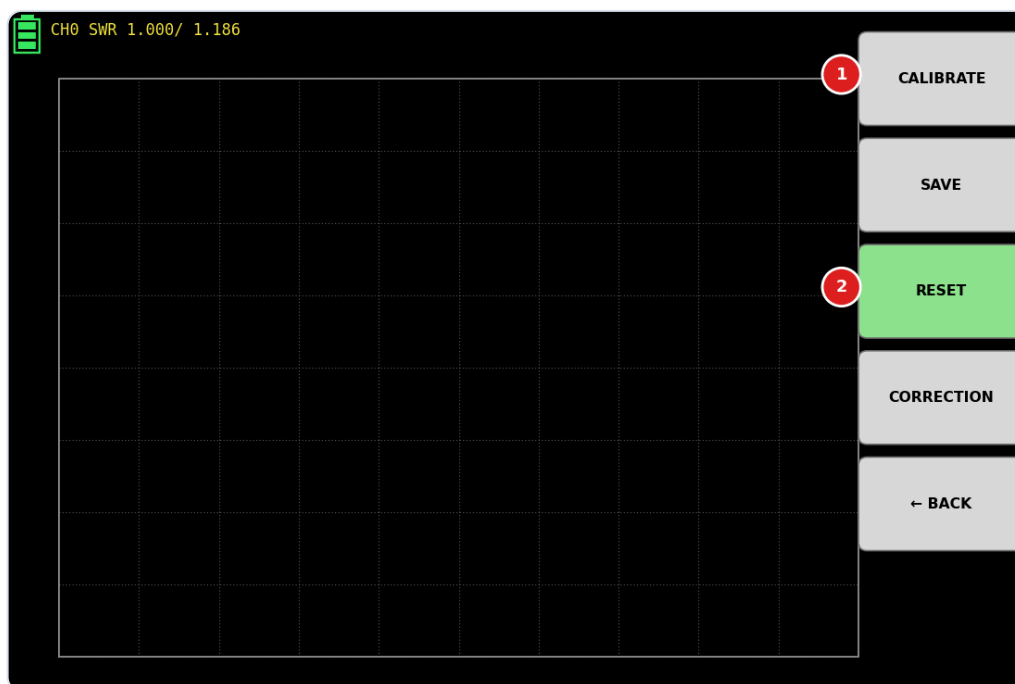


Figure 12 · The CALIBRATE menu.

No.	What you see here
1	CALIBRATE : starts the procedure.
2	RESET : clears the current calibration. Always do this first.

Step 1 First set the range, formats and channel (3.4 and 3.5). The calibration stores those settings along with it.

Step 2 Tap **CALIBRATE > RESET** . The letters on the left of the screen (CO, D, R, S, T, X) disappear.

STEP B: OPEN, SHORT, LOAD



Figure 13 · OPEN is done (black), SHORT is the next step (green).

No.	What you see here
1	Black button: this step is done.
2	Green button: the next step.
3	ISOLN and THRU : only for S21 measurements (filters, chokes). For antennas you skip them.
4	DONE : finish.

Step 1	Tap CALIBRATE > CALIBRATE .
Step 2	Screw the OPEN cap onto the end of the SMA cable with your fingers. Lay the cable down and do not touch it. Wait 2 seconds and tap OPEN .
Step 3	Unscrew the OPEN cap and screw on the SHORT cap. Wait 2 seconds, tap SHORT .
Step 4	Replace it with the LOAD cap. Wait 2 seconds, tap LOAD .
Step 5	Only for filters and chokes: screw the LOAD onto the end of the second cable (the one on CH1) and tap ISOLN . Then join both cable ends with the SMA female-female coupler and tap THRU .
Step 6	Tap DONE .

STEP C: SAVE

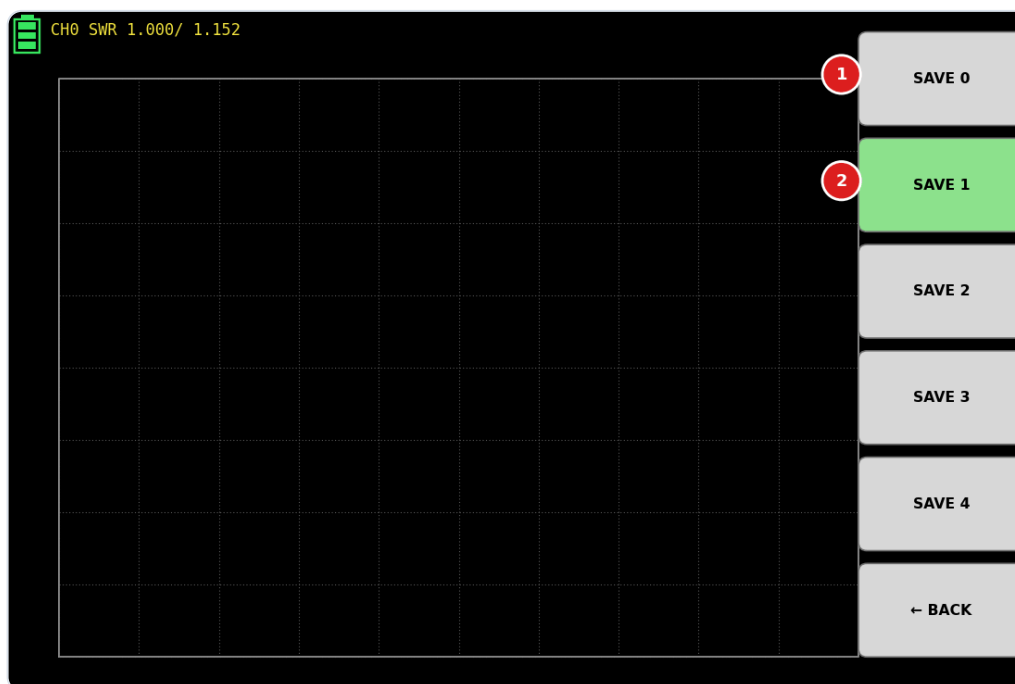


Figure 14 · Choosing a memory slot after DONE.

No.	What you see here
1	SAVE 0 is loaded automatically at power-up. Use it for the measurement you make most often.
2	The selected slot (green). Tap to save.

Step 1 Tap the slot from your memory plan (3.8).

Step 2 The left edge of the screen now shows C with the number of that slot, for example C1.

STEP D: VERIFY

Temporarily set a trace to Smith (**DISPLAY > FORMAT > SMITH**) and screw the three caps on again, one at a time. Each time the dot must land in the right place:

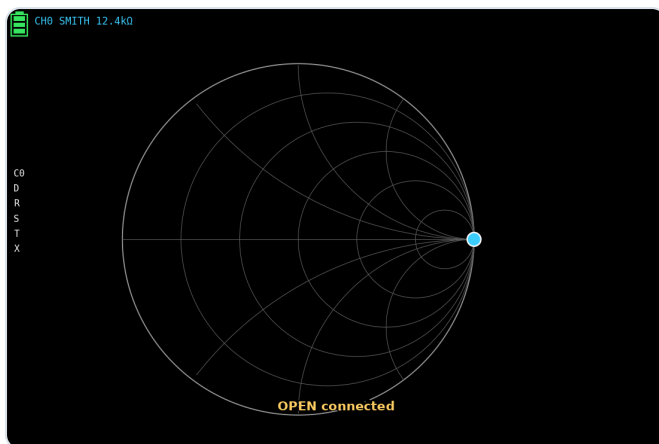


Figure 15 · OPEN: all the way to the right.

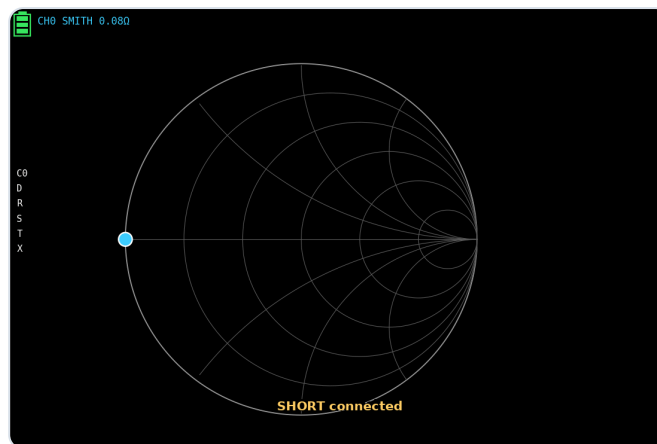


Figure 16 · SHORT: all the way to the left.

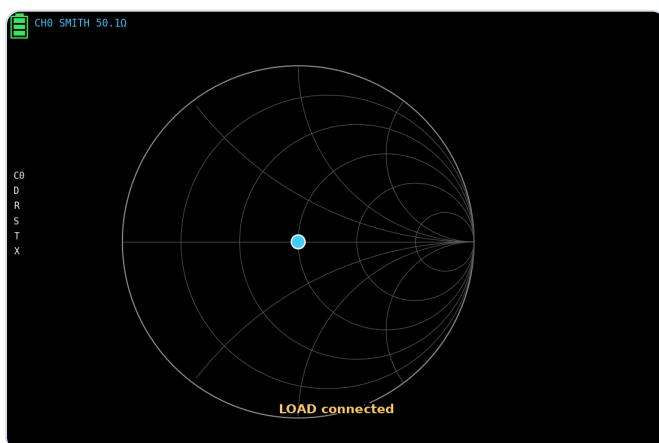


Figure 17 · LOAD: exactly in the centre.

What you see	What it means	What you do
All three in the right place	The calibration is good	Measure
The dot is off, or it jumps around	Cap not tight, cable moved, or wrong order	Tighten the caps, keep the cable still, RESET and start again
The letters C, D, R, S are missing on the left	No calibration is active	Calibrate, or load a saved calibration via RECALL

3.7 Loading a saved calibration

Tap **RECALL > RECALL 1** (or the number you need). Range, formats and calibration all come back. At power-up the NanoVNA automatically loads slot 0.

3.8 Memory plan (a suggestion)

Slot	Range and format	Connection	Use
SAVE 0	6.9 to 7.3 MHz, SWR + Smith	SMA cable + PL	Loaded at power-up
SAVE 1	1 to 30 MHz, SWR	SMA cable + PL	All HF bands
SAVE 2	144 to 146 MHz, SWR	SMA cable + N	2 m antennas

Slot	Range and format	Connection	Use
SAVE 3	1 to 60 MHz, LOGMAG CH1	two SMA cables	Filters
SAVE 4	1 to 30 MHz, LOGMAG CH1	clip leads	Chokes (phase 3)

3.9 Using the marker

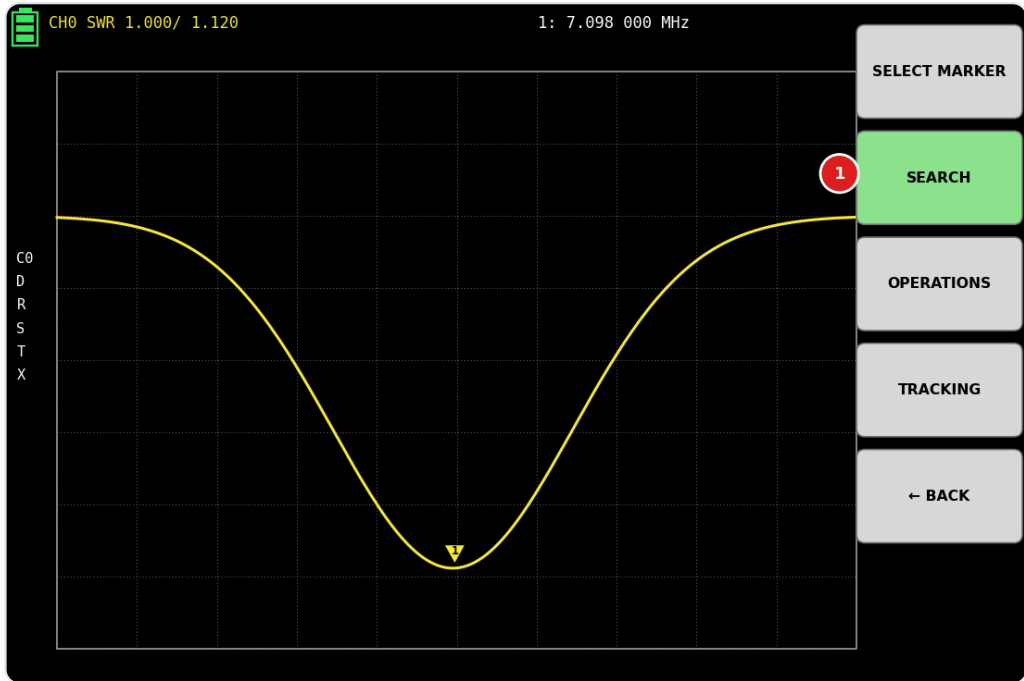


Figure 18 · `MARKER > SEARCH` finds the lowest point for you.

- Step 1** **Manually:** push the jog switch left or right. The marker moves point by point. Or drag the triangle with your stylus.
- Step 2** **Automatically find the lowest SWR:** `MARKER > SEARCH > MINIMUM`. Switch `TRACKING` on and the marker follows the minimum while you work on the antenna.
- Step 3** **A second marker:** `MARKER > SELECT MARKER > MARKER 2`. Handy for marking the start and end of a band.

4. Reading the graphs

The NanoVNA always measures the same thing, but it can present the result in different ways. Each format answers a different question. In this chapter you see **the same 40 m antenna in every format**, so you can compare them directly.

4.1 Which format for which question

Format	Vertical axis	The question it answers	Good
SWR	SWR (1 = perfect)	How well does the antenna accept the power?	Below 1.5 in the band
LOGMAG (on CHO)	Return loss in dB	Same as SWR, but small differences are much easier to see	Deeper than -14 dB
R and X (RESISTANCE, REACTANCE)	Ohm	Where does the antenna resonate, and what is its resistance there?	X = 0 in the band, R close to 50
SMITH	A chart, no axis	Why is the SWR not good: too long, too short, wrong impedance?	Dot close to the centre
PHASE	Degrees	For advanced use, rarely needed for antennas	
LOGMAG (on CH1)	Transmission in dB	How much does a filter or choke attenuate?	Depends on the purpose

4.2 SWR

This is the screen from 3.2. The horizontal axis runs from START to STOP. Vertically you see SWR; with scale 1, each division is 1 SWR unit, and the reference line (the triangle on the left) sits at SWR 1.0.

- Step 1 Find the lowest point of the curve.
- Step 2 Put the marker on it (3.9) and read the frequency and SWR at the top.
- Step 3 Look at how wide the curve stays below SWR 2: that is the usable bandwidth without a tuner.

SWR	Meaning	Action
1.0 to 1.5	Excellent	Nothing to do
1.5 to 2.0	Good; most transceivers deliver full power here	Fine-tune if you like
2.0 to 3.0	Mediocre; the internal tuner is needed	Retune the antenna
Above 3.0	Poor	Check the antenna and the connections

4.3 LOGMAG: return loss

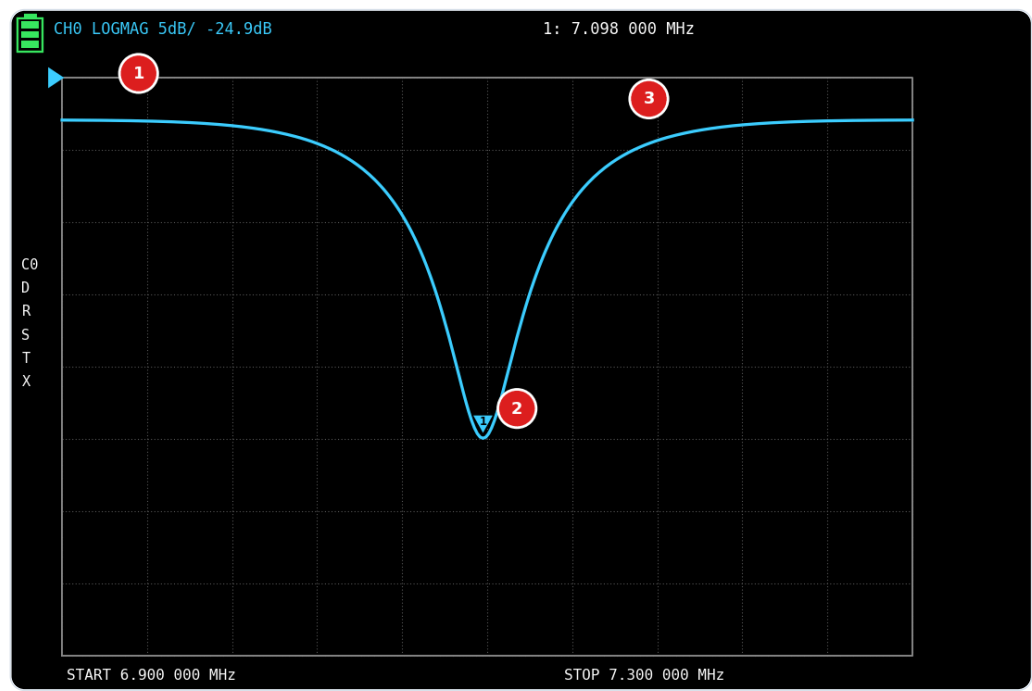


Figure 19 · The same antenna in LOGMAG. The dip is now a deep point pointing down.

No.	What you see here
1	Trace header: LOGMAG, 5 dB per division, value at the marker -24.9 dB.
2	The deepest point = the best match.
3	Reference line at 0 dB: at this level everything is reflected.

LOGMAG on CHO shows the same information as SWR, but in dB. The deeper, the better. The advantage: the difference between *good* and *very good* is far easier to see than on the SWR scale, where everything below 1.2 looks like a flat line.

Return loss	Corresponds to SWR
-6 dB	3.0
-10 dB	1.9
-14 dB	1.5
-20 dB	1.2
-30 dB	1.07

4.4 R and X: resistance and reactance

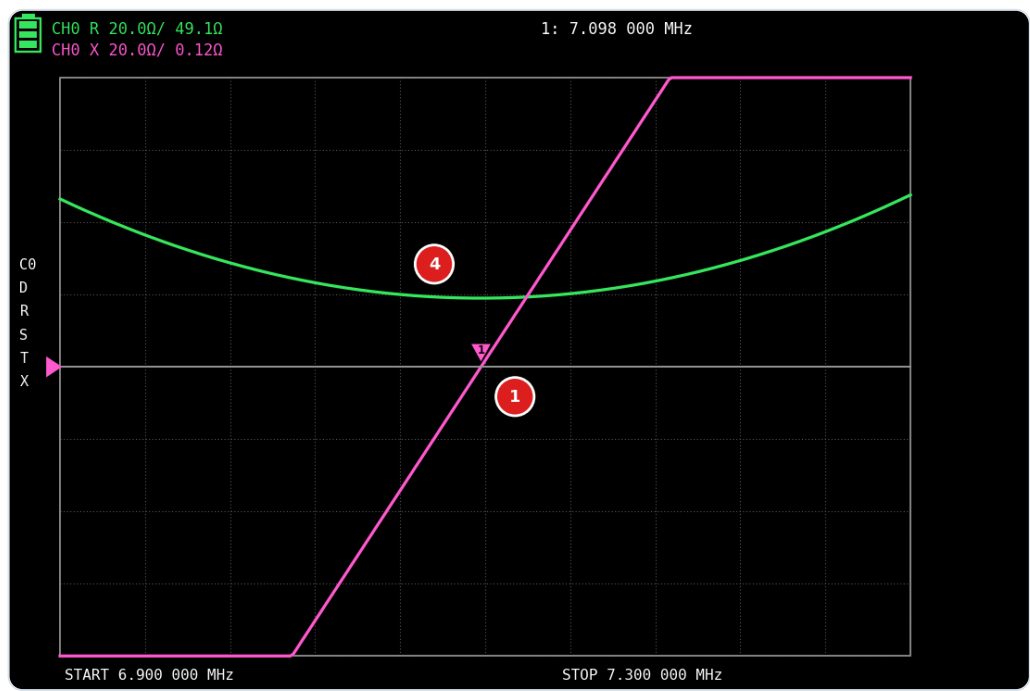


Figure 20 · R (green) and X (pink) of the same antenna.

No.	What you see here	
1	Resonance: here X crosses the zero line. This is the true resonant frequency.	
2	X negative (below the zero line): capacitive, the antenna is too short for this frequency.	
3	X positive (above the zero line): inductive, the antenna is too long for this frequency.	
4	R, the resistance. Read it where X is zero: that is your antenna's impedance at resonance.	

Step 1

Set trace 2 to `FORMAT > MORE > RESISTANCE` and trace 3 to `FORMAT > MORE > REACTANCE` (menu names can differ slightly per firmware), both on CH0, scale 20.

Step 2

Put X's reference line in the middle (`REFERENCE POSITION` 4).

Step 3

Find where X crosses the middle line and read R at that point.

What you see	What it means	What you do
X = 0 in the band, R between 40 and 60 ohm	Resonant and well matched	Nothing
X = 0 in the band, R far from 50 (e.g. 12 or 200 ohm)	Resonant, but the wrong impedance	Different balun or unun ratio
X crosses zero above the band	Antenna too short	Lengthen it
X crosses zero below the band	Antenna too long	Shorten it

4.5 Phase

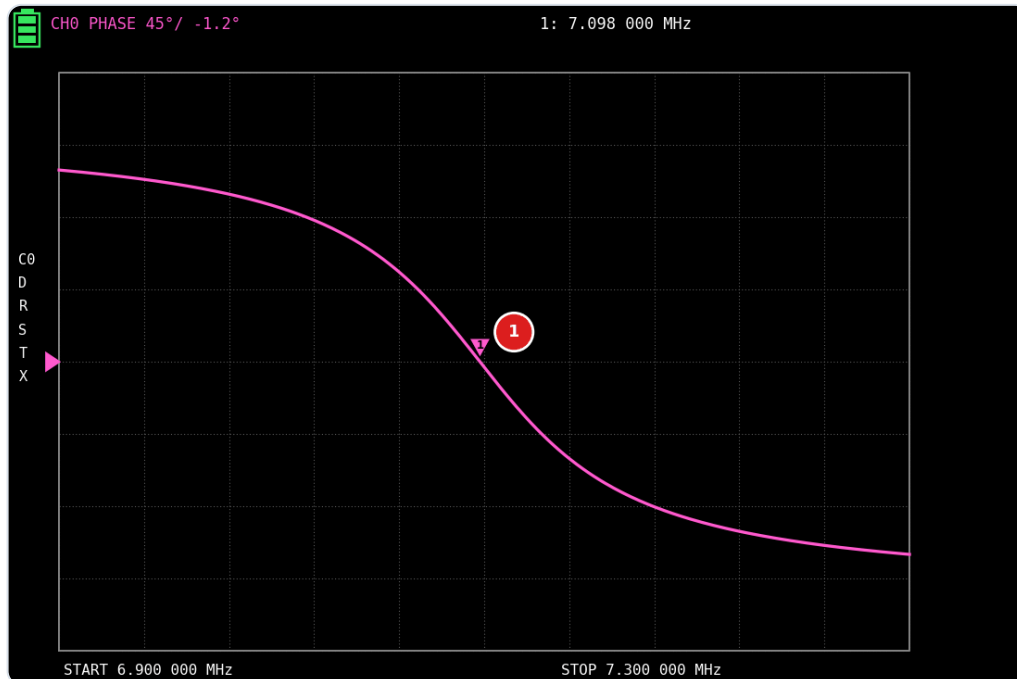


Figure 21 · The phase passes through zero at resonance.

The phase shows whether current and voltage are in step. At resonance it is zero. For antennas it adds nothing beyond R and X or the Smith chart; you can safely ignore it until you start looking at filters or cables in detail.

4.6 Several traces at once

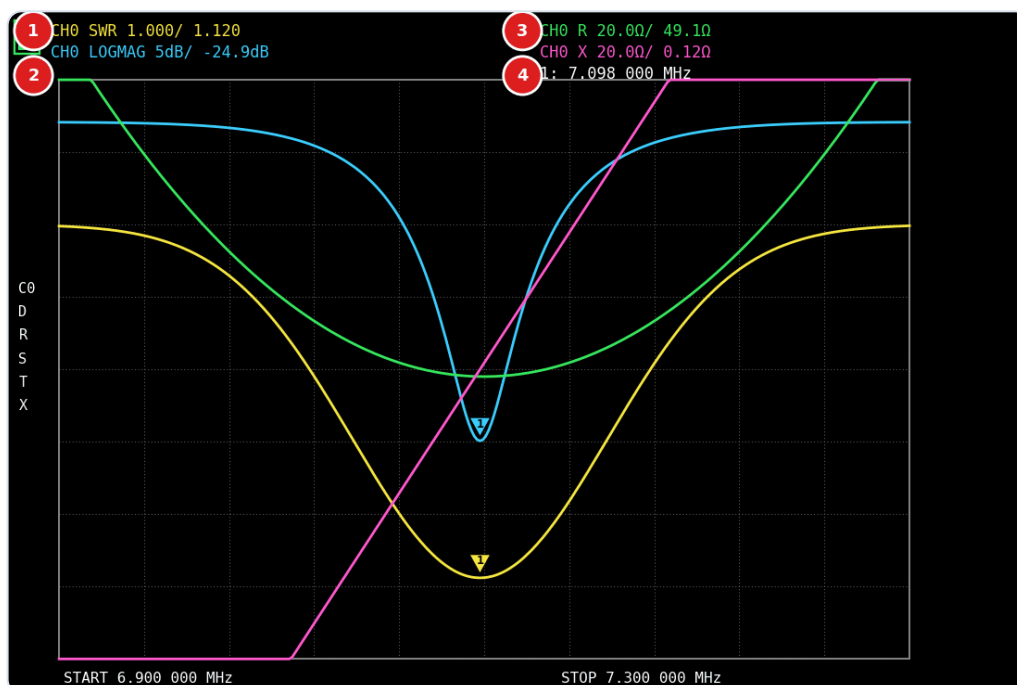


Figure 22 · Four traces at once. Each colour has its own header line at the top.

No.	What you see here
1	Yellow: SWR. The small triangle in front means this is the active trace.

No.	What you see here
2	Blue: LOGMAG (return loss).
3	Green: R , resistance.
4	Pink: X , reactance.

Step 1 Read each header line at the top in the same colour as its line.

Step 2 All traces share the same marker frequency. So for one frequency you see SWR, return loss, R and X together.

Step 3 Too busy? Switch off the traces you do not need via **DISPLAY > TRACE** (3.5).

4.7 LOGMAG on CH1: transmission

For filters and chokes you measure transmission (S21). **0 dB at the top means everything gets through.** The lower the line, the more attenuation. -10 dB is 10 times less power, -20 dB is 100 times less, -30 dB is 1000 times less. Examples in 7.2 and chapter 10.

5. The Smith chart, demystified

5.1 Why a Smith chart?

An SWR graph only tells you *how good* your antenna is. The Smith chart also tells you *why it is not*, and therefore what to do about it. Every point on the chart is an impedance: a combination of resistance (R) and reactance (X). The line you see is how your antenna's impedance changes from the start frequency to the stop frequency.

5.2 The chart, built up in four steps

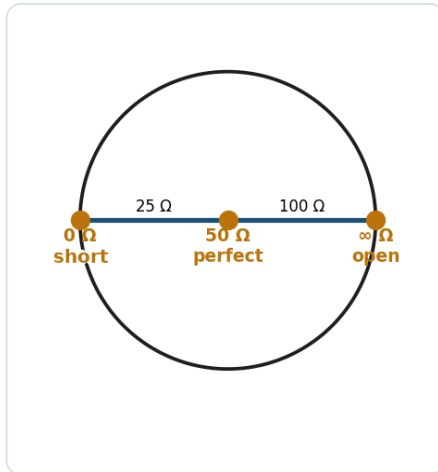


Figure 23 · Step 1: the centre line.

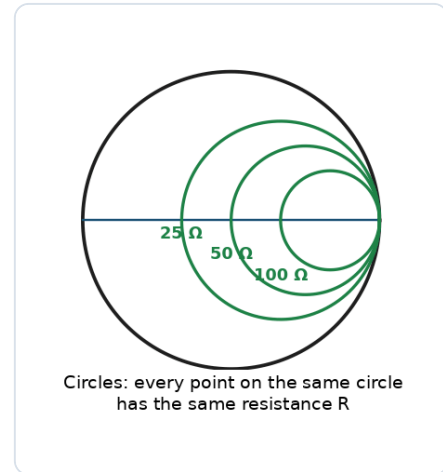


Figure 24 · Step 2: circles of constant resistance.

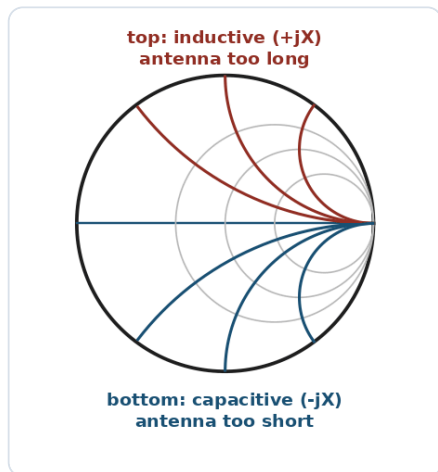


Figure 25 · Step 3: arcs of constant reactance.

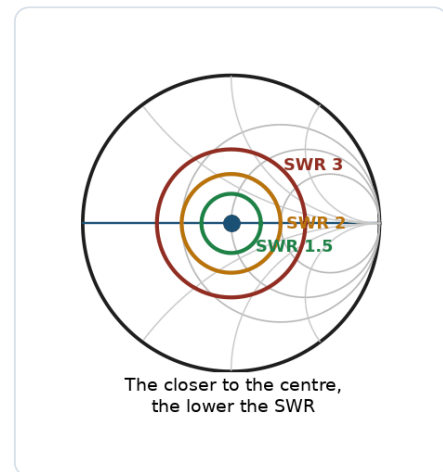


Figure 26 · Step 4: circles of constant SWR.

- Step 1** The horizontal centre line is pure resistance, with no coil or capacitor behaviour. Far left is 0 ohm (a short circuit), the centre is 50 ohm (perfect), far right is infinity (open).
- Step 2** The circles that meet at the right are lines of constant resistance. Everything on the circle through the centre has exactly 50 ohm of resistance.
- Step 3** The arcs above the centre line are inductive (+jX): the antenna behaves like a coil and is too long for that frequency. Below the centre line is capacitive (-jX): too short.

Step 4 **Imagine a circle around the centre point:** everything on that circle has the same SWR. The smaller the circle around the centre, the lower the SWR.

5.3 Reading the marker values at the top

For a Smith trace the top line reads, for example: CH0 SMITH 52.6Ω 205nH. This is how you read it:

Part	Meaning
52.6Ω	The resistance R at the marker frequency: 52.6 ohm.
205nH (nanohenry)	There is also some coil behaviour: inductive, so above the centre line; the antenna is slightly too long.
47.8pF or 47.8nF (picofarad, nanofarad)	Capacitor behaviour: capacitive, below the centre line; the antenna is slightly too short.

TIP

An **H** (henry) in the value means inductive, an **F** (farad) means capacitive. How many ohms the deviation amounts to also depends on the frequency: at 7 MHz, 205 nH is about 9 ohm of reactance. No further maths needed: mainly look at *where* the dot sits.

5.4 Your own antenna on the Smith chart

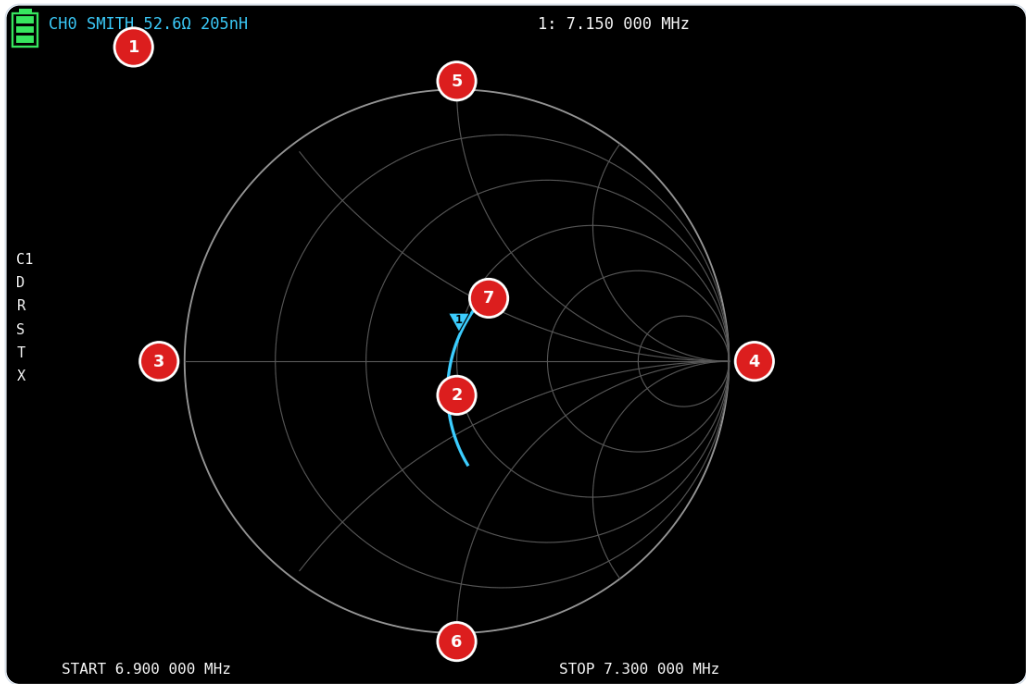


Figure 27 · The 40 m antenna on the Smith chart.

No.	What you see here
1	Trace header: 52.6 ohm and 205 nH at the marker frequency.
2	The centre point: 50 ohm, the target.
3	Left: short circuit.

No.	What you see here
4	Right: open.
5	Lower half: capacitive.
6	Upper half: inductive.
7	Marker 1 at 7.150 MHz.

Step 1 Set a trace to Smith: **DISPLAY > TRACE > TRACE 1**, then **← BACK > FORMAT > SMITH**. Channel CHO REFLECT.

Step 2 Put the marker on the frequency you want to operate on.

Step 3 Look where the dot sits and use the table below.

What you see	What it means	What you do
Dot close to the centre	Good match	Nothing
Dot above the centre line	Inductive: antenna too long for this frequency	Shorten the wire (the dip lies below your frequency)
Dot below the centre line	Capacitive: antenna too short	Lengthen the wire
Line crosses the centre line left of the centre	Resonant, but impedance too low (e.g. 25 ohm)	Different balun/unun ratio
Line crosses right of the centre	Resonant, but impedance too high (e.g. 100 ohm)	Different balun/unun ratio
Small loop around the centre	Well matched over a wide bandwidth	Nothing
Line hugging the outer edge	Nearly everything is reflected: open, short, or a very poor match	Check the connection and the antenna

5.5 Exercise: tuning an antenna tuner with the Smith chart

With a manual antenna tuner you can watch live how the knobs move the dot across the chart. It is the ideal exercise for learning to read a Smith chart.

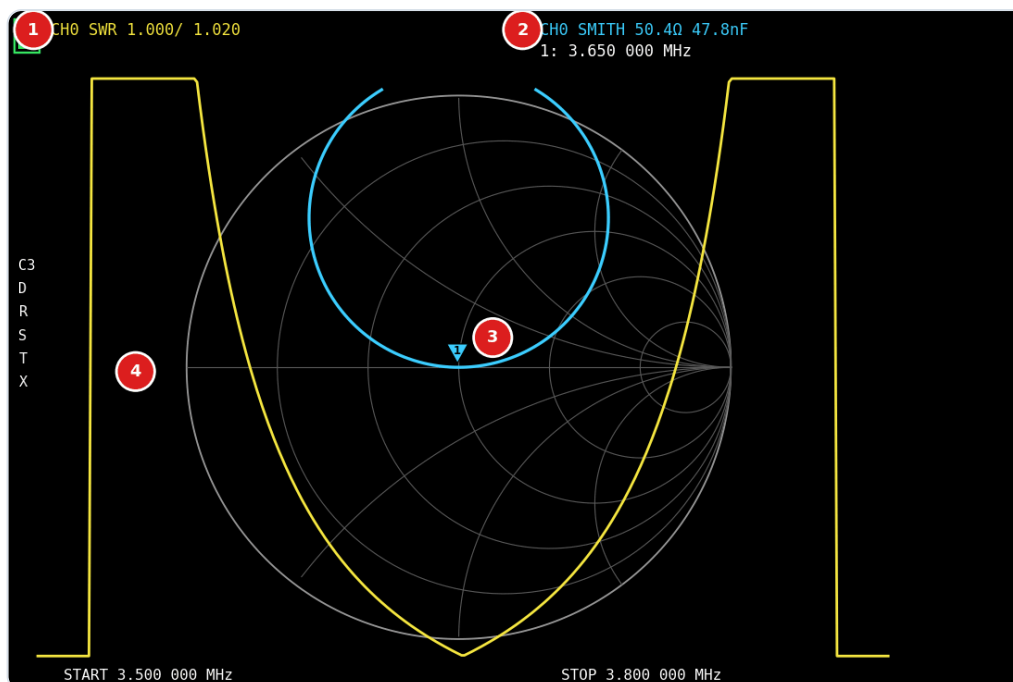


Figure 28 · Tuner perfectly adjusted at 3.650 MHz.

No.	What you see here
1	SWR 1.02 at the marker.
2	50.4 ohm and a negligible reactance: practically perfect.
3	The marker dot sits in the centre of the Smith chart.
4	The yellow SWR curve touches the reference line exactly at the marker frequency.

Step 1 Connect CHO (via the SMA cable and SO-239 adapter) to the tuner's input, where the transceiver normally goes. The antenna stays on the tuner's output.

Step 2 Set START and STOP around the frequency you want (e.g. 3.5 to 3.8 MHz), set trace 0 to SWR and trace 1 to SMITH, and calibrate at the end of the SMA cable.

Step 3 Put the marker on the frequency you want (e.g. 3.650 MHz).

Step 4 Slowly turn one knob of the tuner and watch the dot move across the chart. Then turn the other knob.

Step 5 Stop when the marker dot is in the centre and the SWR at the marker is below 1.2. Write down the knob positions for that band.

6. Phase 1: measuring antennas

6.1 SWR and resonance of an HF antenna

Goal: see at which frequency your antenna works best, and how well. **Example:** an EFHW or end-fed antenna on 40 m.

WHAT YOU NEED

- NanoVNA-H4, the supplied SMA cable, the SMA/SO-239 adapter, the calibration caps, and a wire or shorting plug for discharging.

PROCEDURE

- Step 1 Load your setup:** `RECALL > RECALL 0`. Nothing saved yet? Set 6.9 to 7.3 MHz (3.4), trace 0 to SWR and trace 1 to SMITH, both CHO REFLECT (3.5), and calibrate (3.6).
- Step 2 Disconnect the transceiver** from the antenna switch (0.4) and set the switch to the antenna you want to measure.
- Step 3 Discharge the coax** as described in 0.3: hold the centre pin of the PL-259 against the metal nut for 2 seconds, with a wire or an insulated screwdriver.
- Step 4 Screw the SO-239 adapter** onto the end of the SMA cable, and the coax's PL-259 onto that (2.4).
- Step 5 Tap** `MARKER > SEARCH > MINIMUM`. The marker jumps to the lowest point.
- Step 6 Read** the frequency and SWR of marker 1 at the top and write them down.
- Step 7 Usable bandwidth:** choose `MARKER > SELECT MARKER > MARKER 2` and move it left with the jog switch until the top line shows SWR 2.0. Do the same with marker 3 to the right. The frequency difference between them is the bandwidth below SWR 2.
- Step 8 Finished:** disconnect the VNA and reconnect the transceiver, in reverse order.

HOW TO READ THE RESULT

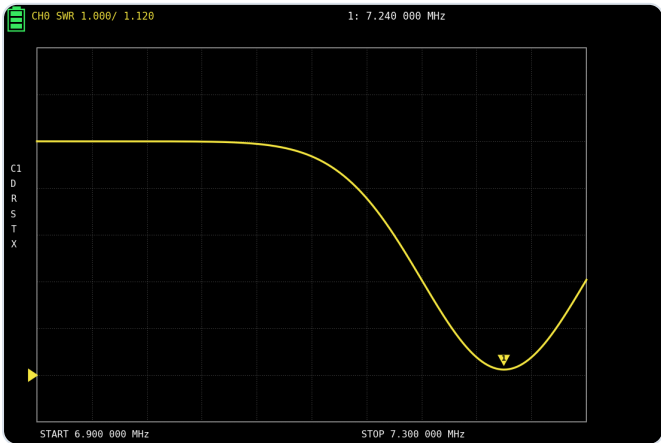


Figure 29 · Dip too high: antenna too short.

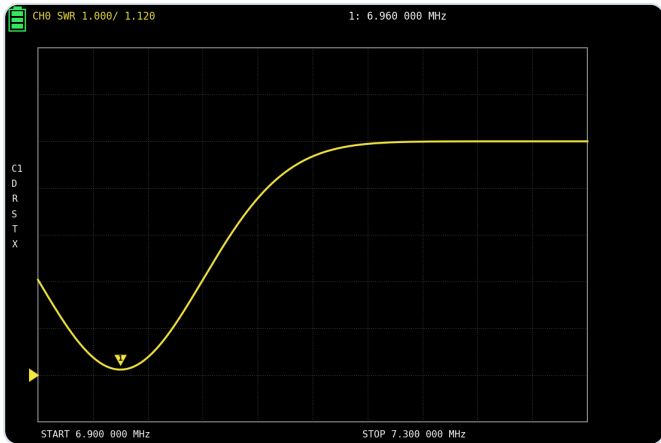


Figure 30 · Dip too low: antenna too long.

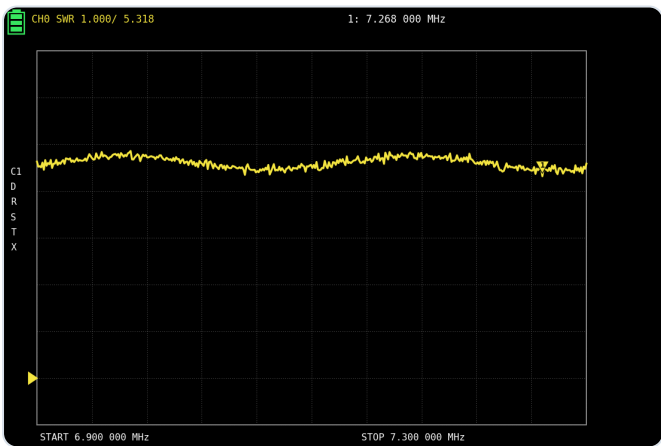


Figure 31 · No dip, high everywhere: fault in the cable or a connection.

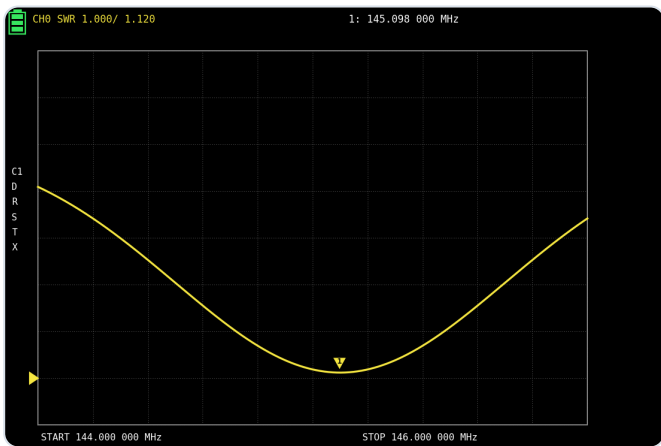


Figure 32 · 2 m antenna via N adapter: dip around 145 MHz.

What you see	What it means	What you do
Dip above the band (e.g. 7.24 MHz)	Antenna electrically too short	Lengthen the wire. Rule of thumb: 1% longer moves the dip about 1% lower
Dip below the band (e.g. 6.96 MHz)	Antenna electrically too long	Shorten the wire in small steps, measuring again each time
Dip in the right place, SWR 1.0 to 1.5	The antenna is good	Nothing; save a measurement as a reference (8.7)
Dip in the right place, lowest SWR above 2	Resonant, but the wrong impedance	Look at the Smith chart (5.4), check the balun/unun
No dip, high and flat everywhere	Open circuit or short circuit	Check connectors and balun, measure the coax on its own (7.1)
Low SWR everywhere, no real dip	Loss in a long coax is hiding the antenna	Measure closer to the antenna (6.4)
Curve keeps jumping	Strong local transmitter or interference	Measure again later, use more points

TIP: SHORTENING WITHOUT CUTTING

Fold the end of the wire back on itself and secure it with insulating tape. That way you can lengthen it again later. Only cut off the surplus once the tuning is final.

6.2 A multiband antenna in one view

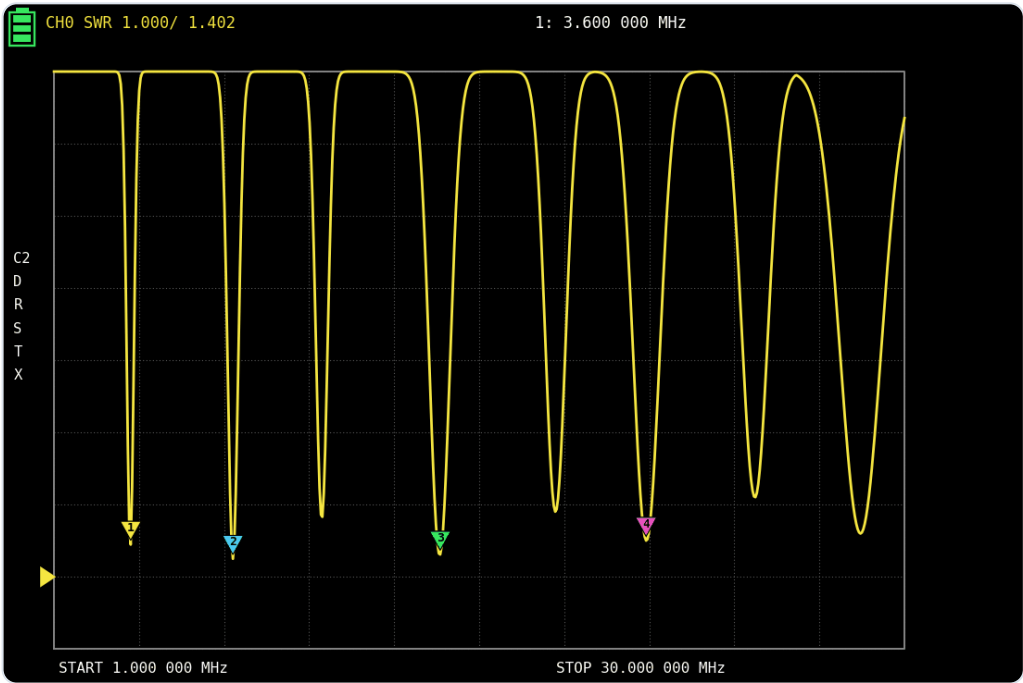


Figure 33 · 1 to 30 MHz. Every dip is a band on which the antenna resonates.

- Step 1 Load `RECALL > RECALL 1`, or set START `1 M`, STOP `30 M` and `SWEEP POINTS > 401`, and calibrate.
- Step 2 Connect the antenna as in 6.1 (discharge first).
- Step 3 Put markers 1 to 4 on the dips of the bands you expect (`MARKER > SELECT MARKER`), then move them with the jog switch).
- Step 4 Is there a dip in every band the antenna is supposed to cover? Then zoom in band by band using the START/STOP values in appendix B.

What you see	What it means	What you do
An expected band without a dip	That band does not work	Zoom in and check the tuning element (coil, loop, link)
Dips outside the amateur bands	Normal for multiband antennas	Only count the dips inside the bands

6.3 Too long or too short: Smith and R/X together

If you are not sure why the SWR is not good, use the Smith chart (5.4) or the R and X view (4.4). Both show whether you need to shorten, lengthen, or change the balun.

6.4 Measuring in the shack or at the antenna

In the shack you measure the **complete system** as your transceiver sees it: antenna, balun, coax, chokes and switch. That is the measurement that counts for your SWR. At the feed point you measure **the antenna without the coax**; you need that to trim the wire length exactly, because coax loss makes the SWR in the shack look better than it is, and the cable length distorts the impedance. The frequency of the dip stays roughly the same.

Step 1 Take the NanoVNA, the SMA cable, the adapter and the caps to the antenna.

Step 2 Calibrate on the spot (3.6).

Step 3 Disconnect the coax from the balun and connect the VNA directly to the balun.

Step 4 Measure as in 6.1.

6.5 VHF and UHF antennas via N

Step 1 Set the range: **STIMULUS** **144 M** to **146 M** for 2 m, or **430 M** to **440 M** for 70 cm.

Step 2 Calibrate at the end of the SMA cable (the caps are SMA).

Step 3 Screw the SMA female-female coupler onto the end of the SMA cable, and the N female/SMA male adapter onto that.

Step 4 Push the N plug of the VHF coax straight onto the adapter and turn the nut clockwise until hand-tight.

Step 5 Save as **SAVE 2** and measure as in 6.1. On 70 cm the adapter can slightly distort the SWR value; the position of the dip remains reliable.

7. Phase 1: coax and filters

7.1 Coax length and fault finding (TDR)

With TDR (Time Domain Reflectometry) the NanoVNA converts the measurement into a distance. An open or shorted end, a bad connector or a crushed cable shows up as a step at a certain distance.



Figure 34 · An open coax of 18.4 m.

No.	What you see here
1	Distance at the marker , calculated with velocity factor 0.66.
2	The step : this is where the cable ends (open). With a short circuit the line goes down instead.

Step 1 Set up: START `50 k`, STOP `900 M`, `SWEEP POINTS > 401`, and calibrate open/short/load at the end of the SMA cable.

Step 2 Connect the coax via the SO-239 adapter. Leave the other end open (nothing connected).

Step 3 Format: `DISPLAY > FORMAT > MORE > REAL`.

Step 4 TDR on: `DISPLAY > TRANSFORM > TRANSFORM ON`, then `LOW PASS STEP`.

Step 5 Velocity factor: `DISPLAY > TRANSFORM > VELOCITY FACTOR`. RG58 and RG213: 66 (or 0.66, depending on firmware). Foam cable such as H155, Aircell 7 or Ecoflex: roughly 80 to 86. If it is not printed on the cable, find the type on the jacket and look it up in the datasheet.

Step 6 Put the marker on the step and read the distance at the top.

What you see	What it means	What you do
Step up at the expected length	Cable fine, end open	Nothing
Step down	Short circuit at that distance	Inspect the connector or cable there
Step much shorter than the cable	Break or damage at that distance	Inspect the cable at that point
Small bump halfway	Poor in-line connector or a kink	Replace the connector there

7.2 Measuring a filter (S21)

Goal: check how much a low-pass filter passes in the band, and how much it attenuates the harmonics. **Example:** a separate 40 m low-pass filter such as a QRP Labs LPF kit.

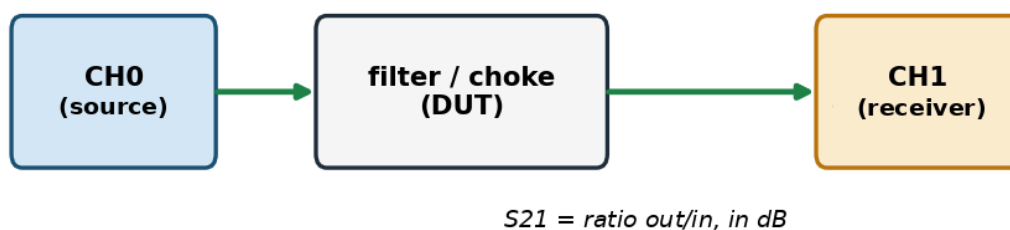


Figure 35 · CH0 sends, the filter sits in between, CH1 receives.

Step 1 Set up: START **1 M**, STOP **60 M**, trace 0 to **LOGMAG** and **CH1 THROUGH**, scale 10, reference 7.

Step 2 Screw the second SMA cable onto CH1.

Step 3 Full calibration: OPEN, SHORT, LOAD at the end of cable 1; LOAD on the end of cable 2 and tap **ISOLN**; join both cable ends with the coupler and tap **THRU**; then **DONE > SAVE 3**.

Step 4 Check: with the cables still joined, the line must sit at 0 dB.

Step 5 Remove the coupler and put the filter between the two cables (if the filter has other connectors, use suitable adapters).

Step 6 Marker 1 on 14.1 MHz (2nd harmonic), marker 2 on 7.05 MHz (operating frequency).

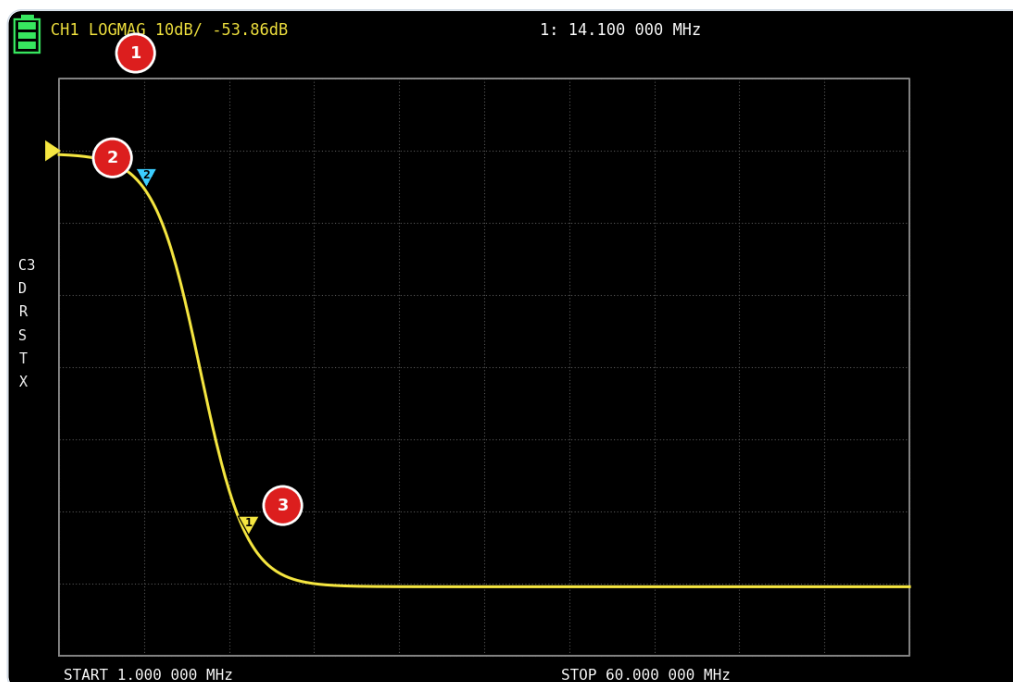


Figure 36 · A 40 m low-pass filter.

No.	What you see here
1	Value at the active marker: -53.9 dB at 14.1 MHz.
2	At 7.05 MHz: almost 0 dB, the signal passes without loss.
3	At 14.1 MHz: more than 50 dB of attenuation on the 2nd harmonic.

What you see	What it means	What you do
Less than 0.5 dB loss at the operating frequency	Good filter	Nothing
More than 1 dB loss at the operating frequency	Filter tuned too low, or a bad component	Check the coils and capacitors
Less than 40 dB attenuation on the 2nd harmonic	Filter insufficient	Different filter or an extra section

8. The NanoVNA on your PC

On the PC you see everything larger, you can look at several graphs at once, use far more measuring points, and save your measurements. The NanoVNA remains the measuring instrument; the PC is just the big screen and the control panel. The free program used here is **NanoVNA-Saver** (Windows, macOS and Linux; the steps below are for Windows).

8.1 Connecting and recognising the NanoVNA

- Step 1** Use the USB-C cable from the box. Some phone cables can only charge and carry no data.
- Step 2** Plug the cable into the NanoVNA and into a USB port on your PC. Switch the NanoVNA on.
- Step 3** Open Device Manager: right-click the Windows Start button and choose **Device Manager**.
- Step 4** Click the arrow in front of **Ports (COM & LPT)**. There is a line **STMicroelectronics Virtual COM Port** with a COM number.
- Step 5** **Check:** unplug the USB cable and the line disappears; plug it back in and the line reappears. Then you have the right port. Remember the COM number.

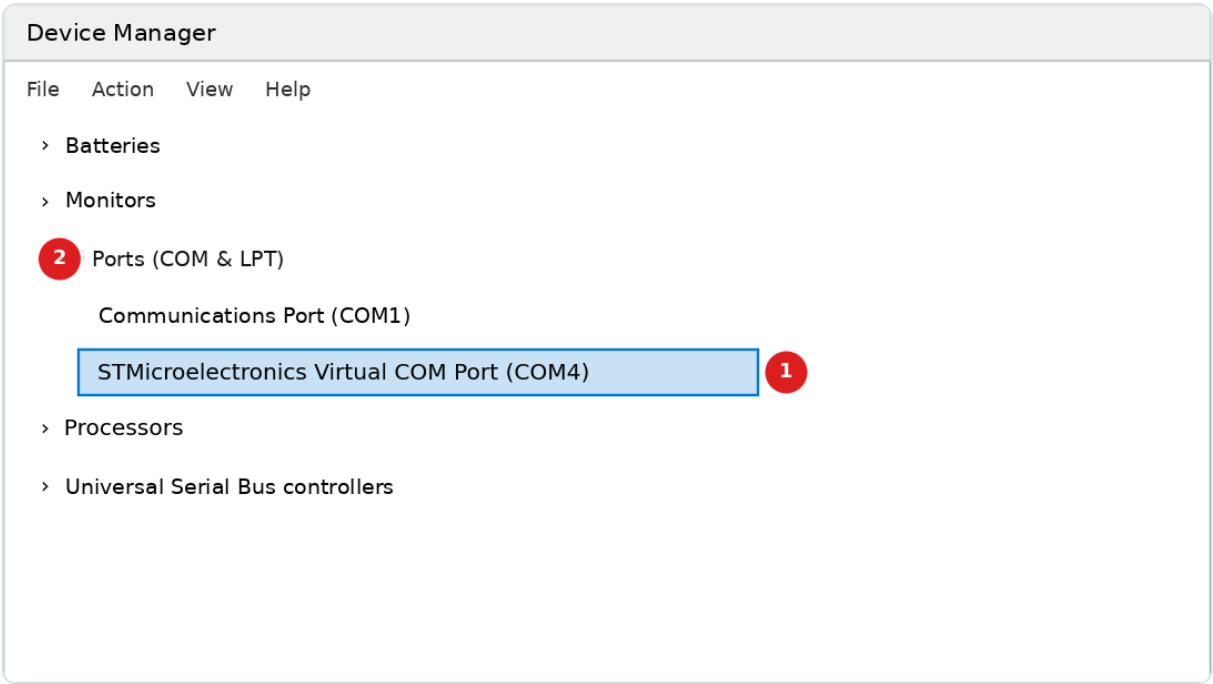


Figure 37 · Device Manager after connecting.

No.	What you see here
1	The NanoVNA. You will choose this COM number (here COM4) in the software.
2	Look under this heading, not under any other.

What you see	What it means	What you do
No new line appears	The cable carries no data, or the driver is missing	Use the cable from the box, try another USB port (preferably USB 2.0), otherwise install the STM32 driver (appendix E)
Yellow exclamation mark	Driver not installed correctly	Right-click the line > Update driver, or reinstall the driver

8.2 Downloading NanoVNA-Saver and connecting

- Step 1** Go to github.com/NanoVNA-Saver/nanovna-saver/releases. At the top (newest) release, open **Assets** and click the Windows **.exe** file. Wait until the download has completely finished.
- Step 2** Double-click the file in your Downloads folder. No installation is needed: NanoVNA-Saver runs straight from the downloaded file. If Windows shows a blue warning, click **More info** and then **Run anyway**.
- Step 3** Bottom left, under **Serial port control**: choose your COM port from the list and click **Connect to device**. The button changes to **Disconnect**.
- Step 4** Top left, under **Sweep control**: enter Start and Stop (e.g. **6.9M** and **7.3M**) and click **Sweep**. After a few seconds the graphs appear on the right: the connection works.

8.3 The window, panel by panel

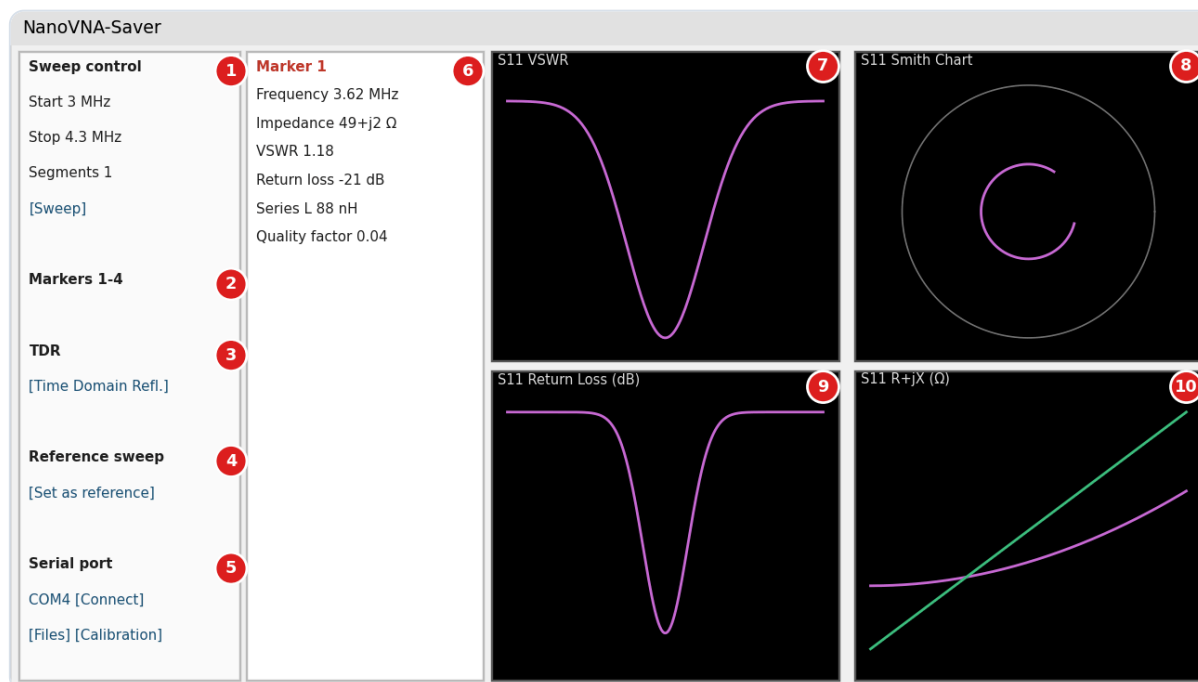


Figure 38 · NanoVNA-Saver with four graphs.

No.	What you see here
1	Sweep control: Start, Stop, Center, Span and Segments. Each segment is an extra sweep of 101 points: 10 segments gives 1010 points, a much finer curve.
2	Markers: up to 4 markers. Type a frequency, or click in a graph to place a marker.

No.	What you see here
3	TDR: opens the window for determining cable length and faults (8.6).
4	Reference sweep: keeps the current measurement as a fixed reference line, to compare before and after (8.5).
5	Serial port control, Files and Calibration: connect, save and calibrate in the software.
6	Marker info: per marker the frequency, impedance ($R + jX$), VSWR, return loss, and the equivalent coil or capacitor.
7	S11 VSWR: the same as SWR on the instrument.
8	S11 Smith Chart: the same as on the instrument, but larger.
9	S11 Return Loss: LOGMAG on CHO (4.3).
10	S11 R+jX: resistance and reactance (4.4).

TIP

You choose which graphs you see via **Display setup** (bottom left). Right-click in a graph to change its scale or type.

8.4 Calibrating in the software

You can use the calibration stored in the instrument itself (then do nothing here), or calibrate in NanoVNA-Saver. The latter has the advantage that the calibration applies to all segments, so a wide multi-segment sweep stays accurate.

Step 1 Click **Calibration**, bottom left.

Step 2 Click **Calibration assistant**. The program asks you to connect open, short and load one at a time; each time, screw the right cap onto the end of the SMA cable and click OK.

Step 3 For filters and chokes, also do through (and isolation) as the assistant asks.

Step 4 Click **Apply**, then **Save calibration** to keep it for next time (use a name with band and connection, e.g. `40m_PL.cal`).

8.5 Comparing before and after

Step 1 Measure your antenna and click **Set current as reference**. The measurement stays on screen as a fixed line.

Step 2 Change the antenna (wire longer or shorter).

Step 3 Click **Sweep**. The new line appears next to the reference, and you see the effect immediately.

Step 4 Click **Reset reference** to clear the reference.

8.6 Cable length in the software (TDR)

Step 1 Set a wide range, e.g. Start **50k**, Stop **900M**, with several segments.

Step 2 Connect the coax with its far end open and click **Sweep**.

Step 3 Click **Time Domain Reflectometry**, then choose the cable type or enter the velocity factor.

Step 4 The window shows the estimated cable length and a graph with the step, as in 7.1.

8.7 Saving for your log or blog

- **Files > Save 1-Port file (S1P):** saves the complete antenna measurement. Via Files you can load it again later to compare.
- **Files > Save 2-Port file (S2P):** for filters and chokes.
- **Screenshot:** press `Windows` + `Shift` + `5` and drag a frame around the window. Paste it into your blog or document.
- **Screenshot of the instrument itself:** use NanoVNA-App by OneOfEleven, which has a button to save the NanoVNA screen as an image.

8.8 Updating the firmware (only if needed)

The firmware on your H4 works fine; you do not need to update to get started. Only do it if you are missing a function or run into a bug. After an update you must redo all calibrations, including TOUCH CAL.

Step 1 Check your current version: `CONFIG > VERSION`.

Step 2 Download the latest DFU firmware for the NanoVNA-H4 (DiSlord) via the nanovna-users group (appendix E). Choose a file that ends in `.dfu` and has **H4** in its name, not just H.

Step 3 Download NanoVNA-App by OneOfEleven, and install the DfuSe software from STMicroelectronics (it contains the DFU driver you need).

Step 4 Put the NanoVNA into DFU mode: `CONFIG > DFU > RESET AND ENTER DFU`. The screen goes black or shows a DFU message. That is normal.

Step 5 Connect it to the PC. Device Manager now shows **STM Device in DFU Mode** under Universal Serial Bus controllers.

Step 6 Open NanoVNA-App, click the **Upload VNA firmware** button, choose your `.dfu` file and wait for the message that the unit must restart.

Step 7 Switch the NanoVNA off and on again. Redo TOUCH CAL (3.1) and all calibrations.



Figure 39 · The NanoVNA in DFU mode.

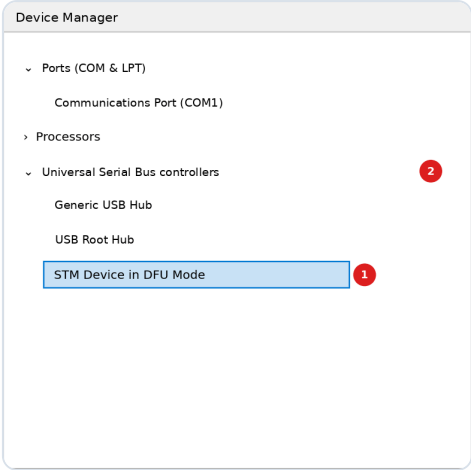


Figure 40 · In DFU mode the unit appears under USB controllers.

No.	What you see here
1	STM Device in DFU Mode: the unit is ready for the update.
2	Look under this heading, no longer under Ports.

CAUTION

Always choose firmware for your model (H4). The wrong firmware will not damage the unit, but it will not work either: repeat the update with the correct file. If you can no longer get into DFU mode, switch the unit off, hold the jog switch pressed in, and switch it on again.

9. Phase 2: the tinySA and your transmitter

9.1 First power-up

- Step 1

Charge the tinySA via USB-C (phone charger or PC) and switch it on.
- Step 2

Calibrate the screen: `CONFIG > TOUCH CAL`, tap the corners it asks for, then `CONFIG > SAVE CONFIG`.
- Step 3

Self test: connect the LOW and HIGH ports with one of the short SMA cables from the box (tighten the nuts with your fingers) and tap `CONFIG > SELF TEST`. The test takes about ten seconds; every line must show PASS.
- Step 4

Level calibration with the same cable: `CONFIG > LEVEL CAL > CALIBRATE`. Then remove the cable.

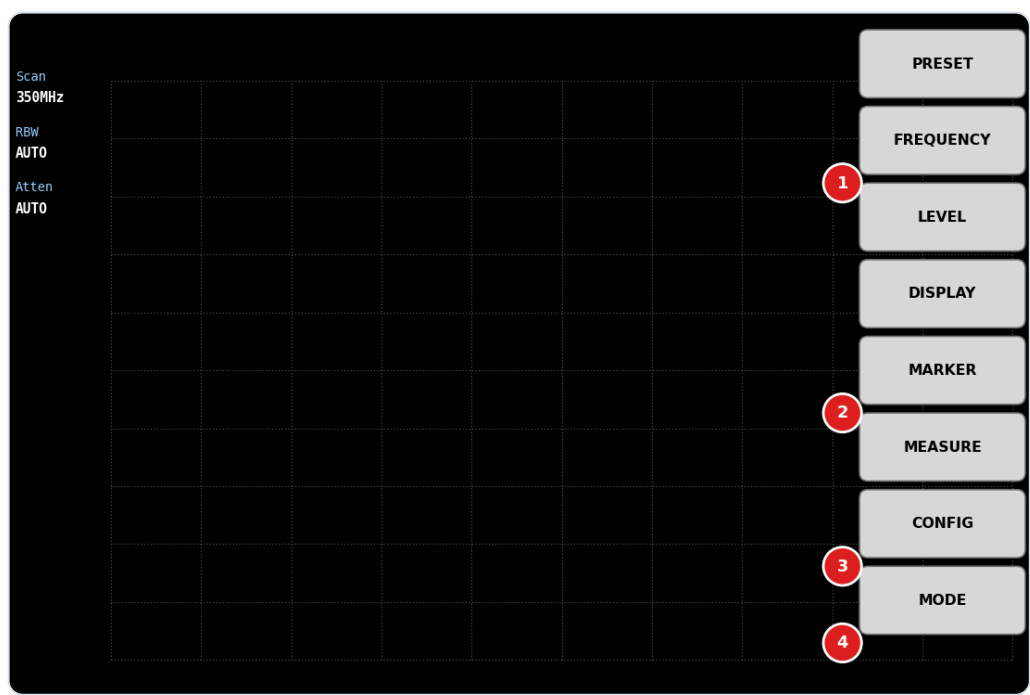


Figure 41 · The tinySA main menu.

No.	What you see here
1	<code>FREQUENCY</code> : start, stop, center, span.
2	<code>MARKER</code> : markers and peak search.
3	<code>CONFIG</code> : self test, calibration, touch cal.
4	<code>MODE</code> : LOW INPUT (HF up to 2 m) or HIGH INPUT (70 cm).

9.2 Why you always use external attenuation

The tinySA's input can take at most **+10 dBm, which is 10 milliwatts**. Your QMX delivers a few watts: hundreds of times more. That is why you **always** put two external attenuators between the transmitter and the tinySA. The internal attenuator is not protection: one wrong setting, or a PRESET that sets it to 0 dB, and the input is gone.

Transmit power	In dBm	After 40 dB	After 40 + 10 dB	Safe?
1 W	+30 dBm	-10 dBm	-20 dBm	Yes
5 W (QMX, IC-7300 at 5 W)	+37 dBm	-3 dBm	-13 dBm	Yes
10 W	+40 dBm	0 dBm	-10 dBm	Yes, but the 40 dB attenuator is at its maximum: only very briefly
100 W	+50 dBm	+10 dBm	0 dBm	NO: the 40 dB attenuator (rated 10 W) burns out

If the dBm column still feels like magic, the [dB & dBm course](#) covers exactly this kind of chain calculation.

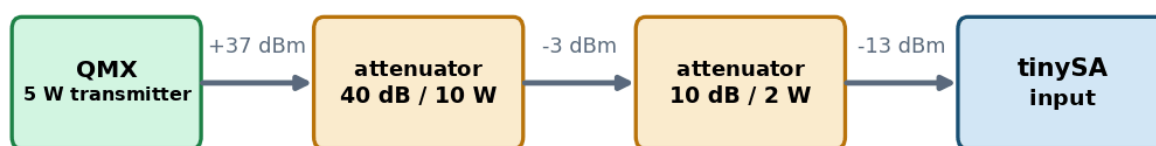
TURNING YOUR TRANSMITTER'S POWER DOWN

- **IC-7300:** press the **MULTI** knob briefly, select **RF POWER** with the knob, and turn it to about 5%. While transmitting, check on the Po meter that you are at around 5 W.
- **QMX:** the output depends on the supply voltage and the version. For the calculation, always assume a maximum of 5 W.

SAFETY

Check the power **before** you connect the attenuators, with the transmitter on a dummy load or on your normal antenna. Never more than 10 W, not even briefly.

9.3 Connecting



Safely below the +10 dBm limit of the tinySA

Figure 42 · The attenuator chain. The 10 W attenuator always comes first.

Step 1 Transmitter **off**.

Step 2 **QMX:** fit the BNC male/SMA female adapter to the BNC antenna socket (push and turn a quarter turn clockwise). **IC-7300:** PL-PL patch cable into the antenna socket, with the SMA/SO-239 adapter on its other end.

Step 3 Screw the **40 dB 10 W attenuator** with its SMA male side onto that adapter. So the side with the heat sink fins is on the transmitter side.

Step 4 Screw the **10 dB 2 W attenuator** with its SMA male side onto the output of the 40 dB one.

Step 5 Connect the output of the 10 dB attenuator to the tinySA's **LOW** input with one of the supplied SMA cables.

Step 6 Check the order one more time: **transmitter, 40 dB, 10 dB, tinySA**.

9.4 Measuring harmonics

SETTING UP

Step 1 **Input:** **MODE > LOW INPUT**.

Step 2 **Range:** **FREQUENCY > START** **0 M**, **FREQUENCY > STOP** **30 M**. For 40 m you then see the fundamental plus the 2nd, 3rd and 4th harmonics.

Step 3 **Account for the external attenuation:** **LEVEL > EXT GAIN**, tap the minus sign, **5**, **0** and **x1**. The tinySA now shows the true level at the transmitter output.

Step 4 **Top of the screen:** **LEVEL > REF LEVEL**, tap **40** and **x1** (+40 dBm, just above 5 W).

Step 5 Look at the status column on the left: it must read **Ext G -50dB**.

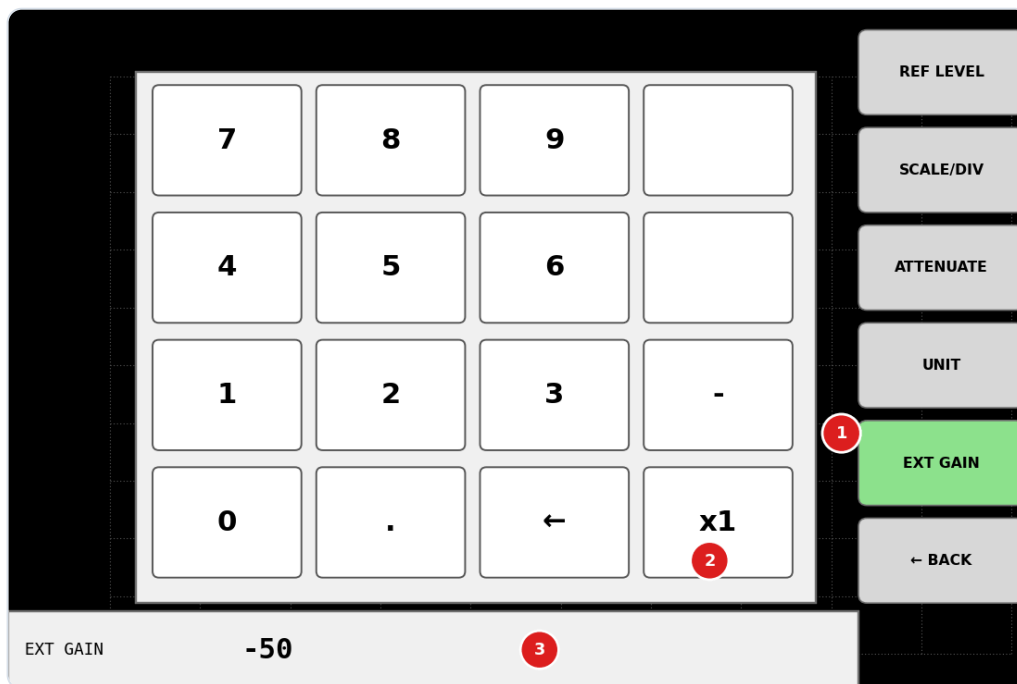


Figure 43 · Setting EXT GAIN to -50 dB.

No.	What you see here
1	The LEVEL menu, with EXT GAIN selected.
2	Minus sign and x1 to confirm.
3	The value entered: -50 (40 + 10 dB of external attenuation).

MEASURING

- Step 1** Set the transmitter to 7.074 MHz, in CW or TUNE (a clean carrier).
- Step 2** Press the CW key or TUNE, count calmly to 5 and release. Then let it rest for at least 20 seconds so the attenuator can cool down.
- Step 3** While transmitting: **MARKER > SEARCH > MAX PEAK** puts marker 1 on the fundamental.
- Step 4** Put marker 2 on 14.148 MHz and marker 3 on 21.222 MHz (twice and three times your frequency). Or use **MEASURE > HARMONIC** if your firmware has it.
- Step 5** Subtract marker 2's level from marker 1's: that is the suppression of the 2nd harmonic in dB, expressed as dBc (dB relative to the carrier).

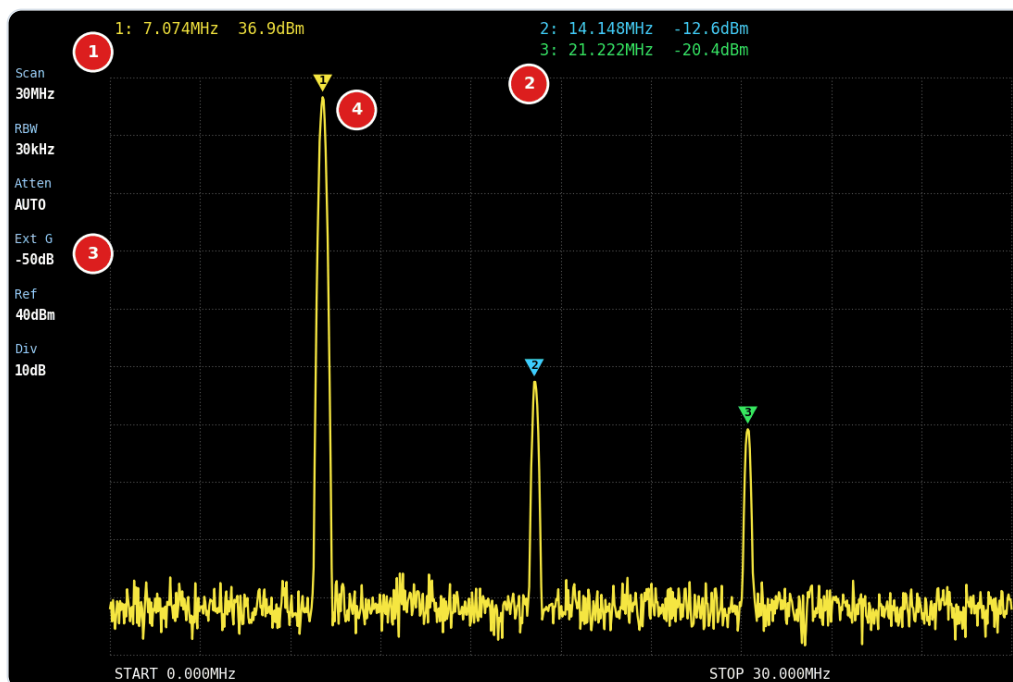


Figure 44 · A QMX on 7.074 MHz with its 2nd and 3rd harmonics.

No. What you see here

- 1** **Marker 1:** fundamental at 7.074 MHz, +36.9 dBm. With EXT GAIN -50 this is the true level: about 4.9 W.
- 2** **Markers 2 and 3:** -12.6 dBm and -20.4 dBm, so 49.5 and 57.3 dB below the fundamental (dBc).
- 3** **Status column:** span, RBW, attenuation, EXT GAIN, reference, scale. Always check for **Ext G -50dB** here.
- 4** **The fundamental:** always the highest peak.

REAL OR FAKE? THE CHECK

A spectrum analyser that receives too much signal generates harmonics of its own. So check your result like this:

- Step 1** Write down how many dB the 2nd harmonic is below the fundamental (e.g. 49.5 dB).
- Step 2** Transmitter off. Unscrew the 10 dB attenuator and connect the 40 dB attenuator directly to the tinySA (5 W then becomes -3 dBm, still safe). Set EXT GAIN to -40.

Step 3 Measure again and compare.

What you see	What it means	What you do
The difference between fundamental and harmonic stays the same	The harmonic really comes from your transmitter	The result is reliable
The difference changes clearly	The tinySA is generating part of it	Always measure with both attenuators and use that value

JUDGING THE RESULT

What you see	What it means	What you do
2nd and 3rd harmonic more than 50 dB below the fundamental	A very clean signal	Nothing
Between 40 and 50 dB	Acceptable for QRP; check the current BIPT requirements	Consider an extra low-pass filter
Less than 40 dB	Too much unwanted emission	Do not transmit on that band; check the filter (7.2)

For a worked example of what a full spectrum analysis of a real transmitter looks like, see [my Yaesu FT-65 under the microscope](#).

9.5 Checking frequency and power

- **Frequency:** marker 1 with MAX PEAK. For a precise reading, narrow the range, e.g. 7.0 to 7.15 MHz.
- **Power:** with EXT GAIN -50 you read the level directly in dBm: +37 dBm is 5 W, +33 dBm is 2 W, +30 dBm is 1 W. Accuracy is about 1 to 2 dB.

9.6 Hunting interference sources

Here you transmit nothing: the tinySA is completely safe and works as a sensitive receiver. You do not need attenuators for this.

Step 1 Set **LEVEL > EXT GAIN** back to 0.

Step 2 Screw the supplied telescopic antenna (or a 1 m piece of wire) onto the LOW input.

Step 3 Set the range to the band with the interference: **FREQUENCY > START** **6.9 M** and **STOP** **7.3 M**.

Step 4 Walk through the house with the tinySA. Where the peaks are highest, the source is close.

Step 5 Switch suspect devices off one at a time and watch whether the peaks disappear.

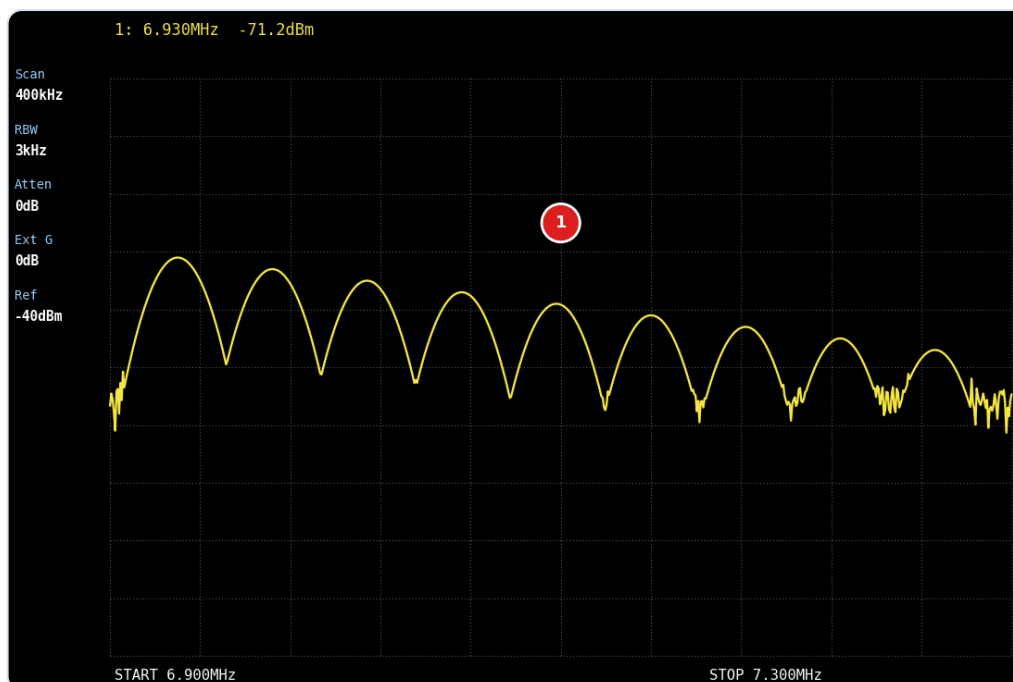


Figure 45 · Interference with a regular pattern of peaks.

No.	What you see here	
1	Peaks at equal spacing: typical of switch-mode power supplies, chargers and inverters.	
What you see	What it means	What you do
Peaks at a fixed spacing	Switch-mode power supply or inverter	Switch devices off one at a time
A broad rise in the noise floor	Broadband interference (PLC, LED lighting)	The same, plus ferrites on the power leads (chapter 10)

10. Phase 3: chokes and ferrites

10.1 What are you measuring?

A choke lets the normal signal *inside* the coax pass, but blocks current flowing over the *outside* of the shield: the **common-mode current**. That current carries RF into the house. The higher the choke's impedance for that current, the better.

CAUTION

The ordinary S21 measurement from 7.2 (through the connectors, through the coax) shows almost 0 dB with a good choke and tells you nothing about choking. The test signal has to run over the *outside* of the shield. That is what the clip leads are for.

10.2 Connecting

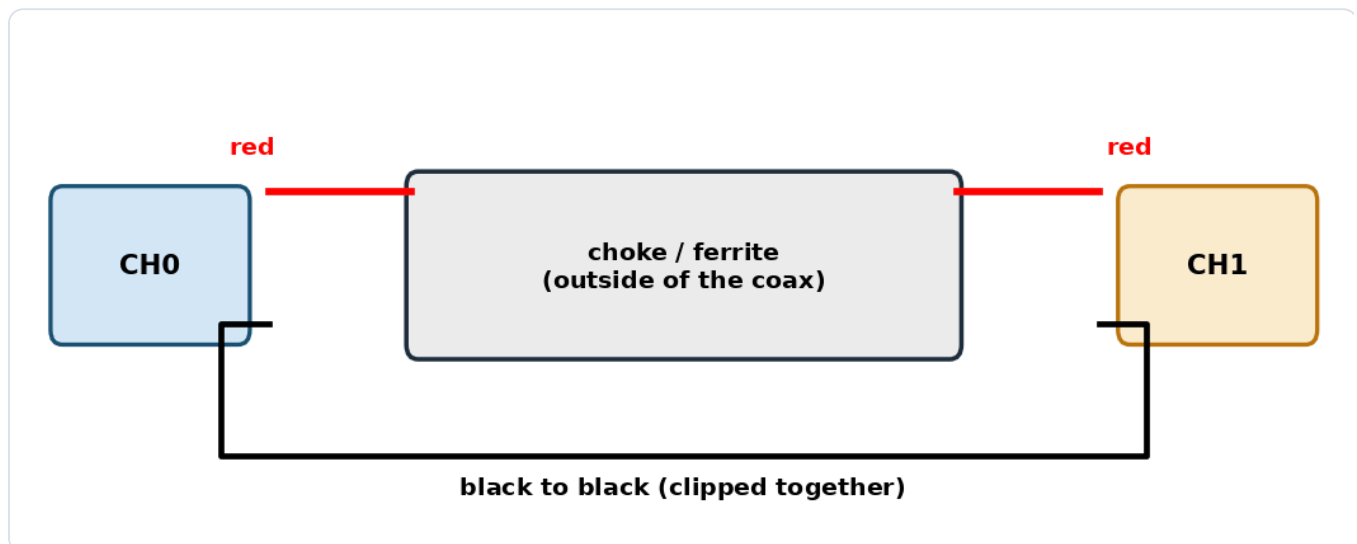


Figure 46 · Red clips on the metal body of the connectors on both sides, black clips joined together.

- Step 1** Screw a BNC female/SMA male adapter onto CHO and onto CH1 (turn only the small nut, with your fingers).
- Step 2** Push the BNC plug of each clip lead onto an adapter and turn a quarter turn clockwise until it clicks.
- Step 3** Clip the red clip from CHO onto the metal outside (the nut) of the connector on one side of the choke. **Never on the centre pin.**
- Step 4** Clip the red clip from CH1 onto the metal outside of the connector on the other side.
- Step 5** Clip the two black clips to each other, and keep them away from the choke.

10.3 Calibrating and measuring

- Step 1** **Set up:** START **1 M**, STOP **30 M**, **LOGMAG**, **CH1 THROUGH**, scale 10, reference 7.

Step 2 Calibration without the choke: clip red to red and black to black. Tap **CALIBRATE > RESET**, then **CALIBRATE > CALIBRATE > THRU**, then **DONE > SAVE 4**. The line must now sit at 0 dB.

Step 3 Clip the choke in between, as in 10.2.

Step 4 Put marker 1 on 7.1 MHz (or the band with the interference) and read the attenuation at the top.

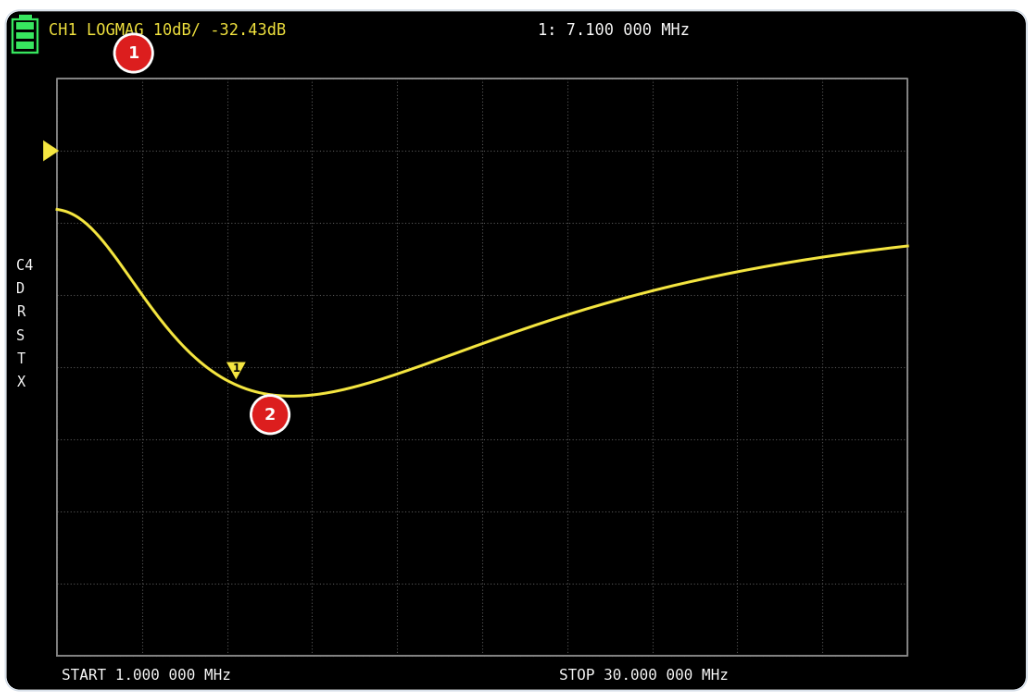


Figure 47 · A good choke around 40 m.

No.	What you see here
1	Attenuation at the marker: about -32 dB at 7.1 MHz.
2	The marker on the band with the interference.

10.4 Converting to ohms

The choke's impedance is roughly $Z \approx 100 \times (10^{\text{attenuation}/20} - 1)$ ohm. No maths? Use this table.

Attenuation	Impedance	Verdict
10 dB	about 220 ohm	Weak
20 dB	about 900 ohm	Minimum
26 dB	about 1900 ohm	Good
30 dB	about 3100 ohm	Very good
40 dB	about 9900 ohm	Excellent

What you see	What it means	What you do
Less than 20 dB on the band with interference	The choke is too weak on that band	More turns, or a different ferrite material (e.g. mix 31 for HF)

What you see	What it means	What you do
High attenuation on a different band	The choke is not tuned to your band	A different choke, or a different number of turns
More than 26 dB on the band with interference	A good choke	Nothing

If you want to go deeper into ferrite materials, my post on [ferrite mix 43 and the 49:1 unun](#) compares mixes with bench measurements.

10.5 Measuring clip-on ferrites on a cable

Step 1 Take a piece of wire about 50 cm long, or the cable the ferrite is meant for.

Step 2 Snap the ferrite shut around the wire.

Step 3 Red clips on the two ends of the wire, black clips joined together.

Step 4 Measure with 1 ferrite, then 2, then 3 side by side, and note the attenuation at 7.1 MHz each time.

Step 5 This shows you how much each extra ferrite really adds.

Appendices

A. Quick card: measuring an antenna

Step 1 **RECALL > RECALL 0** (or set up and calibrate, 3.6).

Step 2 Disconnect the transceiver (0.4).

Step 3 Discharge the coax: centre pin against the nut for 2 seconds (0.3).

Step 4 Connect the coax via the SO-239 adapter.

Step 5 **MARKER > SEARCH > MINIMUM**.

Step 6 Read the frequency and SWR, compare with the table in 6.1.

Step 7 Disconnect the VNA, reconnect the transceiver.

B. Start and stop per band

Band	START	STOP
160 m	1.8 M	2.0 M
80 m	3.5 M	3.8 M
60 m	5.3 M	5.4 M
40 m	6.9 M	7.3 M
30 m	10.0 M	10.2 M
20 m	13.9 M	14.5 M
17 m	18.0 M	18.2 M
15 m	20.9 M	21.5 M
12 m	24.8 M	25.0 M
10 m	28.0 M	29.7 M
6 m	50.0 M	52.0 M
2 m	144 M	146 M
70 cm	430 M	440 M

C. Conversion tables

SWR	Return loss	Reflected power
1.0	infinite	0%
1.2	-20.8 dB	0.8%

SWR	Return loss	Reflected power
1.5	-14.0 dB	4%
2.0	-9.5 dB	11%
3.0	-6.0 dB	25%

Power	dBm
1 mW	0 dBm
10 mW	+10 dBm (tinySA limit)
1 W	+30 dBm
5 W	+37 dBm
10 W	+40 dBm
100 W	+50 dBm

D. Glossary

Term	Meaning
VNA	Vector network analyser: measures reflection and transmission
SWR	Standing wave ratio: how well the antenna accepts the power
S11	Reflection at port 1 (CHO)
S21	Transmission from port 1 to port 2
Return loss	S11 in dB: the deeper, the better
R and X	Resistance and reactance, together the impedance
Inductive / capacitive	Coil or capacitor behaviour; for antennas: too long or too short
OSL	Open, short, load: the three calibration caps
THRU	A direct connection, used for S21 calibration
Smith chart	A chart showing resistance and reactance together
TDR	Conversion to distance: cable length and faults
DFU	Mode for loading new firmware
dBm	Power relative to 1 milliwatt
dBc	Level relative to the carrier (the fundamental)
EXT GAIN	tinySA correction for external attenuation
Common mode	Current flowing over the outside of the coax shield

E. Links and sources

- [tinySA wiki \(official\)](#)
- [NanoVNA-Saver releases](#)
- [nanovna-users group \(firmware, manuals\)](#)
- [NanoVNA-App \(OneOfEleven\)](#)
- [STM32 Virtual COM Port driver](#)
- [DfuSe \(DFU driver and software\)](#)
- [W2AEW's videos on the NanoVNA](#), an excellent visual companion to this guide

Background reading: *Absolute Beginner's Guide to NanoVNA* by Martin Svaco 9A2JK, and the *NanoVNA User Guide* by Hugen. Both are well worth reading once the basics in this guide feel comfortable.

F. Menu names per firmware

Depending on the firmware on your unit, a menu may have a slightly different name. This table helps you find the right button.

In this guide	Older firmware (edy555, Hugen)	Note
CALIBRATE	CAL	Same menu
CALIBRATE > CALIBRATE	CAL > CALIBRATE	Open, short, load, isoln, thru, done
RECALL	RECALL or RECALL/SAVE	Same function
STIMULUS > SWEEP POINTS	not present	Older firmware always measures 101 points
MARKER > SEARCH	not present	Move the marker manually
DISPLAY > TRANSFORM	sometimes not present	Use TDR in NanoVNA-Saver instead (8.6)
FORMAT > MORE > RESISTANCE / REACTANCE	FORMAT > MORE	Names may be abbreviated (R, X)